Satisficing 2: Post-LCA | Revision

anonymized

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cmLC | classification (modal) to LC, based on estimated posterior probabilities estpostprob | estimated posterior probabilities

0.1 Get data

```
col_names_wave1_cawi <- c(

    "EXT1",
    "MID1",
    "FTM1",
    "INR1",
    "NDWO1",
    "NDSS1",
    "NDSO1",
    "BO1",
    "SPEE1",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
```

```
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"



)
file_path_wave1_cawi <-
  "C:[redacted]
wave1_cawi <- read.table(file_path_wave1_cawi, header = FALSE, col.names =
  col_names_wave1_cawi)

wave1_cawi <-
  wave1_cawi |>
  rename(pid = PID,
         hid = HID,
         cprob_opt_wave1 = CPROB2,
         cprob_indiff_wave1 = CPROB1,
         cprob_exm_wave1 = CPROB3) |>
  mutate(
    LC_wave1 = case_when(
      C == 2 ~ "Softsat",
      C == 1 ~ "Indifferent",
      C == 3 ~ "Extremist"
    ),
    LC_wave1 = factor(LC_wave1, levels = c("Softsat", "Indifferent", "Extremist")),
    Mode_wave1 = "Web"
  )




col_names_w2_cawi <- c(
    "EXT2",
    "MID2",
    "FTM2",
    "INR2",
    "NDWO2",
    "NDSS2",
    "NDSO2",
    "BO2",
```

```
    "SPEE2",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)
file_path_w2_cawi <-
  "C:/[redacted]
w2_cawi <- read.table(file_path_w2_cawi, header = FALSE, col.names = col_names_w2_cawi)

w2_cawi <-
  w2_cawi |>
  rename(pid = PID,
         hid = HID,
         cprob_opt_w2 = CPROB1,
         cprob_indiff_w2 = CPROB3,
         cprob_exm_w2 = CPROB2) |>
  mutate(
    LC_w2 = case_when(
      C == 1 ~ "Softsat",
      C == 3 ~ "Indifferent",
      C == 2 ~ "Extremist"
    ),
    LC_w2 = factor(LC_w2, levels = c("Softsat", "Indifferent", "Extremist")),
    Mode_w2 = "Web"
  )




col_names_wave3_cawi <- c(
    "EXT3",
    "MID3",
    "FTM3",
    "INR3",
    "NDWO3",
    "NDSS3",
```

```
    "NDSO3",
    "BO3",
    "SPEE3",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)
file_path_wave3_cawi <-
  "C:[redacted]
wave3_cawi <- read.table(file_path_wave3_cawi, header = FALSE, col.names =
  col_names_wave3_cawi)

wave3_cawi <-
  wave3_cawi |>
  rename(pid = PID,
         hid = HID,
         cprob_exm_wave3 = CPROB1,
         cprob_indiff_wave3 = CPROB2,
         cprob_opt_wave3 = CPROB3) |>
  mutate(
    LC_wave3 = case_when(
      C == 1 ~ "Extremist",
      C == 2 ~ "Indifferent",
      C == 3 ~ "Softsat"
    ),
    LC_wave3 = factor(LC_wave3, levels = c("Softsat", "Indifferent", "Extremist")),
    Mode_wave3 = "Web"
  )



col_names_wave1_papi <- c(
  "EXT1",
    "MID1",
    "FTM1",
    "INR1",
```

```
    "NDWO1",
    "NDSS1",
    "NDSO1",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)
file_path_wave1_papi <-
  "C:/[redacted]
wave1_papi <- read.table(file_path_wave1_papi, header = FALSE, col.names =
  col_names_wave1_papi)

wave1_papi <-
  wave1_papi |>
  rename(pid = PID,
         hid = HID,
         cprob_indiff_wave1 = CPROB1,
         cprob_opt_wave1 = CPROB3,
         cprob_exm_wave1 = CPROB2) |>
  mutate(
    LC_wave1 = case_when(
      C == 2 ~ "Extremist",
      C == 1 ~ "Indifferent",
      C == 3 ~ "Softsat"
    ),
    LC_wave1 = factor(LC_wave1, c("Softsat", "Indifferent", "Extremist")),
    Mode_wave1 = "Paper"
  )


col_names_w2_papi <- c(
  "EXT2",
    "MID2",
    "FTM2",
    "INR2",
    "NDWO2",
    "NDSS2",
```

```
    "NDSO2",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)
file_path_w2_papi <-
  "C:/[redacted]
w2_papi <- read.table(file_path_w2_papi, header = FALSE, col.names = col_names_w2_papi)

w2_papi <-
  w2_papi |>
  rename(pid = PID,
         hid = HID,
         cprob_indiff_w2 = CPROB1,
         cprob_opt_w2 = CPROB3,
         cprob_exm_w2 = CPROB2) |>
  mutate(
    LC_w2 = case_when(
      C == 2 - "Extremist",
      C == 1 - "Indifferent",
      C == 3 - "Softsat"
    ),
    LC_w2 = factor(LC_w2, c("Softsat", "Indifferent", "Extremist")),
    Mode_w2 = "Paper"
  )


col_names_wave3_papi <- c(
    "EXT3",
    "MID3",
    "FTM3",
    "INR3",
    "NDWO3",
    "NDSS3",
    "NDSO3",
    "PID",
    "TGANC",
```

```
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)
file_path_wave3_papi <-
  "C:/[redacted]
wave3_papi <- read.table(file_path_wave3_papi, header = FALSE, col.names =
  col_names_wave3_papi)

wave3_papi <-
  wave3_papi |>
  rename(pid = PID,
         hid = HID,
         cprob_indiff_wave3 = CPROB1,
         cprob_opt_wave3 = CPROB3,
         cprob_exm_wave3 = CPROB2) |>
  mutate(
    LC_wave3 = case_when(
      C == 2 - "Extremist",
      C == 1 - "Indifferent",
      C == 3 - "Softsat"
    ),
    LC_wave1 = factor(LC_wave3, c("Softsat", "Indifferent", "Extremist")),
    Mode_wave3 = "Paper"
  )


wave1_cawi |> ai()
```

```
Rows: 6
Columns: 31
$ EXT1              <dbl> 0, 0, 0, 0, 1, 0
$ MID1              <dbl> 0, 0, 0, 0, 0, 0
$ FTM1              <dbl> 0, 0, 0, 0, 0, 0
$ INR1              <dbl> 0, 0, 0, 0, 0, 1
$ NDWO1             <dbl> 1, 0, 0, 1, 0, 0
$ NDSS1             <dbl> 1, 1, 0, 1, 0, 1
$ NDSO1             <dbl> 0, 0, 1, 0, 0, 0
$ BO1               <dbl> 0, 0, 0, 0, 0, 0
```

```
$ SPEE1              <dbl> 0, 0, 0, 0, 0, 0
$ TGANC              <dbl> 2, 2, 2, 2, 2, 2
$ AGET1              <dbl> 1, 1, 2, 1, 2, 2
$ AGET3              <dbl> 1, 2, 1, 1, 1, 1
$ GENF               <dbl> 1, 2, 1, 1, 1, 1
$ MIGY               <dbl> 2, 1, 1, 2, 1, 1
$ EDTERT             <dbl> 2, 2, 1, 1, 2, 1
$ EDNA               <dbl> 1, 1, 1, 1, 1, 1
$ EMPP               <dbl> 1, 1, 1, 2, 1, 1
$ EMPNO              <dbl> 1, 1, 1, 1, 1, 1
$ EMPRET             <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA              <dbl> 1, 1, 1, 1, 1, 1
$ KIDY               <dbl> 2, 1, 2, 1, 2, 1
$ KIDNA              <dbl> 1, 1, 1, 1, 1, 1
$ INCT1              <dbl> 2, 1, 1, 2, 1, 2
$ INCT3              <dbl> 1, 2, 1, 1, 2, 1
$ INCNA              <dbl> 1, 1, 1, 1, 1, 1
$ cprob_indiff_wave1 <dbl> 0, 0, 1, 0, 0, 0
$ cprob_opt_wave1    <dbl> 0.981, 0.988, 0.000, 0.981, 0.981, 0.696
$ cprob_exm_wave1    <dbl> 0.019, 0.012, 0.000, 0.019, 0.019, 0.304
$ C                  <dbl> 2, 2, 1, 2, 2, 2
$ LC_wave1           <fct> Softsat, Softsat, Indifferent, Softsat, Softsat, So~
$ Mode_wave1         <chr> "Web", "Web", "Web", "Web", "Web", "Web"
```

```
wave1_papi |> ai()
```

```
Rows: 6
Columns: 29
$ EXT1               <dbl> 0, 0, 0, 0, 0, 0
$ MID1               <dbl> 0, 0, 0, 0, 0, 0
$ FTM1               <dbl> 0, 0, 1, 1, 0, 0
$ INR1               <dbl> 0, 0, 0, 1, 0, 0
$ NDWO1              <dbl> 0, 0, 0, 1, 0, 1
$ NDSS1              <dbl> 0, 1, 1, 0, 0, 1
$ NDSO1              <dbl> 0, 0, 0, 1, 1, 0
$ TGANC              <dbl> 2, 2, 2, 2, 2, 2
$ AGET1              <dbl> 2, 2, 2, 1, 2, 1
$ AGET3              <dbl> 1, 1, 1, 1, 1, 1
$ GENF               <dbl> 1, 2, 1, 1, 2, 1
$ MIGY               <dbl> 1, 1, 2, 1, 1, 1
$ EDTERT             <dbl> 1, 1, 2, 1, 1, 2
$ EDNA               <dbl> 1, 1, 1, 1, 1, 1
$ EMPP               <dbl> 2, 1, 1, 1, 1, 1
$ EMPNO              <dbl> 1, 2, 1, 1, 1, 1
$ EMPRET             <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA              <dbl> 1, 1, 1, 1, 1, 1
$ KIDY               <dbl> 1, 1, 2, 1, 2, 1
$ KIDNA              <dbl> 1, 1, 1, 1, 1, 1
$ INCT1              <dbl> 2, 1, 1, 1, 1, 2
$ INCT3              <dbl> 1, 1, 1, 1, 1, 1
$ INCNA              <dbl> 1, 1, 2, 2, 1, 1
$ cprob_indiff_wave1 <dbl> 0.024, 0.000, 0.000, 1.000, 1.000, 0.000
$ cprob_exm_wave1    <dbl> 0.040, 0.025, 0.330, 0.000, 0.000, 0.019
$ cprob_opt_wave1    <dbl> 0.936, 0.975, 0.670, 0.000, 0.000, 0.981
$ C                  <dbl> 3, 3, 3, 1, 1, 3
$ LC_wave1           <fct> Softsat, Softsat, Softsat, Indifferent, Indifferent~
$ Mode_wave1         <chr> "Paper", "Paper", "Paper", "Paper", "Paper", "Paper"
```

```r
socdem <- c(
  "TGANC", "AGET1", "AGET3", "GENF", "MIGY", "EDTERT", "EDNA", "EMPP", "EMPNO", "EDNA",
   ↪  "EMPRET", "EMPNA", "KIDY", "KIDNA", "INCT1", "INCT3", "INCT3","INCNA"
)

wave1 <- rbind(wave1_cawi |> select(pid, hid, Mode_wave1, LC_wave1, starts_with("cprob_"),
 ↪  all_of(socdem)),
               wave1_papi |> select(pid, hid, Mode_wave1, LC_wave1, starts_with("cprob_"),
 ↪  all_of(socdem)))
w2 <- rbind(w2_cawi |> select(pid, hid, Mode_w2, LC_w2, starts_with("cprob_")),
               w2_papi |> select(pid, hid, Mode_w2, LC_w2, starts_with("cprob_")))
wave3 <- rbind(wave3_cawi |> select(pid, hid, Mode_wave3, LC_wave3, starts_with("cprob_")),
               wave3_papi |> select(pid, hid, Mode_wave3, LC_wave3, starts_with("cprob_")))

df <-
  wave1 |>
  left_join(w2, by = c("pid", "hid")) |>
  left_join(wave3, by = c("pid", "hid")) |>
  mutate(
    Age_cat = case_when(
      AGET1 == 2 ~ "Age_T1[18.0-40.0]",
      AGET3 == 2 ~ "Age_T3[60.0-97.0]",
      TRUE       ~ "Age_T2[40.0-60.0]"
    ) |> factor(levels = c("Age_T1[18.0-40.0]", "Age_T2[40.0-60.0]", "Age_T3[60.0-97.0]")),

    Mig = case_when(
      MIGY == 2 ~ "Yes",
      TRUE      ~ "No"
    ) |> factor(levels = c("No", "Yes")),

    Gender = case_when(
      GENF == 2 ~ "Female",
      TRUE      ~ "Not female"
    ) |> factor(levels = c("Not female", "Female")),

    Education = case_when(
      EDTERT == 2 ~ "Tertiary (ISCED 5-8)",
      EDNA   == 2 ~ "missing",
      TRUE        ~ "Non-tertiary (ISCED 0-4)"
    ) |> factor(levels = c("Non-tertiary (ISCED 0-4)", "Tertiary (ISCED 5-8)", "missing")),

    Employment = case_when(
      EMPP   == 2 ~ "Part-time or flexible",
      EMPNO  == 2 ~ "Not currently employed, training",
      EMPRET == 2 ~ "Retirement",
      EMPNA  == 2 ~ "missing",
      TRUE        ~ "Full-time"
    ) |> factor(levels = c("Full-time", "Part-time or flexible",
                           "Not currently employed, training", "Retirement", "missing")),

    ChildHH = case_when(
      KIDY  == 2 ~ "Child in hh",
      KIDNA == 2 ~ "missing",
      TRUE       ~ "No child"
    ) |> factor(levels = c("No child", "Child in hh", "missing")),
```

```
    EquivIncome = case_when(
      INCT1 == 2 ~ "EquivIncome_T1min",
      INCT3 == 2 ~ "EquivIncome_T3max",
      INCNA == 2 ~ "missing",
      TRUE       ~ "EquivIncome_T2med"
    ) |> factor(levels = c("EquivIncome_T1min", "EquivIncome_T2med",
                           "EquivIncome_T3max", "missing")),

    TargetGroup = case_when(
      TGANC == 2 ~ "Anchor",
      TRUE       ~ "HH member"
    ) |> factor(levels = c("HH member", "Anchor"))
  )

df |> ai()
```

```
Rows: 6
Columns: 39
$ Mode_wave1         <chr> "Web", "Web", "Web", "Web", "Web", "Web"
$ LC_wave1           <fct> Softsat, Softsat, Indifferent, Softsat, Softsat, So~
$ cprob_indiff_wave1 <dbl> 0, 0, 1, 0, 0, 0
$ cprob_opt_wave1    <dbl> 0.981, 0.988, 0.000, 0.981, 0.981, 0.696
$ cprob_exm_wave1    <dbl> 0.019, 0.012, 0.000, 0.019, 0.019, 0.304
$ TGANC              <dbl> 2, 2, 2, 2, 2, 2
$ AGET1              <dbl> 1, 1, 2, 1, 2, 2
$ AGET3              <dbl> 1, 2, 1, 1, 1, 1
$ GENF               <dbl> 1, 2, 1, 1, 1, 1
$ MIGY               <dbl> 2, 1, 1, 2, 1, 1
$ EDTERT             <dbl> 2, 2, 1, 1, 2, 1
$ EDNA               <dbl> 1, 1, 1, 1, 1, 1
$ EMPP               <dbl> 1, 1, 1, 2, 1, 1
$ EMPNO              <dbl> 1, 1, 1, 1, 1, 1
$ EMPRET             <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA              <dbl> 1, 1, 1, 1, 1, 1
$ KIDY               <dbl> 2, 1, 2, 1, 2, 1
$ KIDNA              <dbl> 1, 1, 1, 1, 1, 1
$ INCT1              <dbl> 2, 1, 1, 2, 1, 2
$ INCT3              <dbl> 1, 2, 1, 1, 2, 1
$ INCNA              <dbl> 1, 1, 1, 1, 1, 1
$ Mode_w2            <chr> "Web", "Web", "Web", "Web", "Web", "Web"
$ LC_w2              <fct> Softsat, Indifferent, Softsat, Indifferent, Softsat~
$ cprob_opt_w2       <dbl> 0.998, 0.000, 0.727, 0.000, 0.998, 0.013
$ cprob_exm_w2       <dbl> 0.002, 0.000, 0.273, 0.000, 0.002, 0.987
$ cprob_indiff_w2    <dbl> 0, 1, 0, 1, 0, 0
$ Mode_wave3         <chr> "Web", "Web", "Web", "Web", "Web", "Web"
$ LC_wave3           <fct> Indifferent, Indifferent, Extremist, Indifferent, S~
$ cprob_exm_wave3    <dbl> 0.00, 0.00, 0.72, 0.00, 0.00, 0.00
$ cprob_indiff_wave3 <dbl> 1, 1, 0, 1, 0, 1
$ cprob_opt_wave3    <dbl> 0.00, 0.00, 0.28, 0.00, 1.00, 0.00
$ Age_cat            <fct> Age_T2[40.0-60.0], Age_T3[60.0-97.0], Age_T1[18.0-4~
$ Mig                <fct> Yes, No, No, Yes, No, No
$ Gender             <fct> Not female, Female, Not female, Not female, Not fem~
$ Education          <fct> Tertiary (ISCED 5-8), Tertiary (ISCED 5-8), Non-ter~
$ Employment         <fct> "Full-time", "Full-time", "Full-time", "Part-time o~
$ ChildHH            <fct> Child in hh, No child, Child in hh, No child, Child~
$ EquivIncome        <fct> EquivIncome_T1min, EquivIncome_T3max, EquivIncome_T~
```

```
$ TargetGroup        <fct> Anchor, Anchor, Anchor, Anchor, Anchor, Anchor
```

```
df |> filter(LC_wave1 == "Extremist") |> group_by(Mode_wave1) |> frq(EquivIncome)
```

```
EquivIncome <categorical>
# grouped by: Paper
# total N=459 valid N=459 mean=2.82 sd=1.28

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min | 118 | 25.71 |   25.71 |  25.71
EquivIncome_T2med |  67 | 14.60 |   14.60 |  40.31
EquivIncome_T3max |  52 | 11.33 |   11.33 |  51.63
missing           | 222 | 48.37 |   48.37 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>

EquivIncome <categorical>
# grouped by: Web
# total N=193 valid N=193 mean=3.18 sd=1.12

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min |  29 | 15.03 |   15.03 |  15.03
EquivIncome_T2med |  20 | 10.36 |   10.36 |  25.39
EquivIncome_T3max |  32 | 16.58 |   16.58 |  41.97
missing           | 112 | 58.03 |   58.03 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>
```

```
df |> filter(LC_w2 == "Extremist") |> group_by(Mode_w2) |> frq(EquivIncome)
```

```
EquivIncome <categorical>
# grouped by: Paper
# total N=749 valid N=749 mean=2.34 sd=1.22

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min | 263 | 35.11 |   35.11 |  35.11
EquivIncome_T2med | 182 | 24.30 |   24.30 |  59.41
EquivIncome_T3max |  92 | 12.28 |   12.28 |  71.70
missing           | 212 | 28.30 |   28.30 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>

EquivIncome <categorical>
# grouped by: Web
# total N=286 valid N=286 mean=2.81 sd=1.23

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min |  67 | 23.43 |   23.43 |  23.43
EquivIncome_T2med |  48 | 16.78 |   16.78 |  40.21
EquivIncome_T3max |  44 | 15.38 |   15.38 |  55.59
missing           | 127 | 44.41 |   44.41 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>
```

```
df |> filter(LC_wave3 == "Extremist") |> group_by(Mode_wave3) |> frq(EquivIncome)
```

```
EquivIncome <categorical>
# grouped by: Paper
```

```
# total N=308 valid N=308 mean=2.27 sd=1.25

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min | 123 | 39.94 |   39.94 |  39.94
EquivIncome_T2med |  64 | 20.78 |   20.78 |  60.71
EquivIncome_T3max |  35 | 11.36 |   11.36 |  72.08
missing           |  86 | 27.92 |   27.92 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>

EquivIncome <categorical>
# grouped by: Web
# total N=1323 valid N=1323 mean=2.45 sd=1.05

Value             |   N | Raw % | Valid % | Cum. %
--------------------------------------------------
EquivIncome_T1min | 312 | 23.58 |   23.58 |  23.58
EquivIncome_T2med | 351 | 26.53 |   26.53 |  50.11
EquivIncome_T3max | 412 | 31.14 |   31.14 |  81.25
missing           | 248 | 18.75 |   18.75 | 100.00
<NA>              |   0 |  0.00 |    <NA> |   <NA>
```

1 Load SensAn

```
col_names_wave1_papisensan <- c(
  "EXT1",
    "MID1",
    "FTM1",
    "INR1",
    "NDWO1",
    "NDSS1",
    "NDSO1",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "PCONS",
    "CPROB1",
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
```

```
)
file_path_wave1_papisensan <-
  "C:/[redacted]
wave1_papi_sensan <- read.table(file_path_wave1_papisensan, header = FALSE, col.names =
  col_names_wave1_papisensan)

wave1_papi_sensan <-
  wave1_papi_sensan |>
  rename(pid = PID,
         hid = HID,
         cprob_indiff = CPROB1,
         cprob_opt = CPROB3,
         cprob_3 = CPROB2) |>
  mutate(
    LC = case_when(
      C == 2 ~ "Extremist",
      C == 1 ~ "Indifferent",
      C == 3 ~ "Softsat"
    ),
    LC = factor(LC, c("Softsat", "Indifferent", "Extremist")),
    Mode = "Paper"
  )



col_names_wave1_cawisensan <- c(

    "EXT1",
    "MID1",
    "FTM1",
    "INR1",
    "NDWO1",
    "NDSS1",
    "NDSO1",
    "BO1",
    "SPEE1",
    "PID",
    "TGANC",
    "AGET1",
    "AGET3",
    "GENF",
    "MIGY",
    "EDTERT",
    "EDNA",
    "EMPP",
    "EMPNO",
    "EMPRET",
    "EMPNA",
    "KIDY",
    "KIDNA",
    "INCT1",
    "INCT3",
    "INCNA",
    "PCONS",
    "CPROB1",
```

```
    "CPROB2",
    "CPROB3",
    "C",
    "HID"
)


file_path_wave1_cawisensan <-
  "C:/[redacted]
wave1_cawi_sensan <- read.table(file_path_wave1_cawisensan, header = FALSE, col.names =
  col_names_wave1_cawisensan)

wave1_cawi_sensan <-
  wave1_cawi_sensan |>
  rename(pid = PID,
         hid = HID,
         cprob_opt = CPROB2,
         cprob_indiff = CPROB1,
         cprob_3 = CPROB3) |>
  mutate(
    LC = case_when(
      C == 2 - "Softsat",
      C == 1 - "Indifferent",
      C == 3 - "Complex"
    ),
    LC = factor(LC, levels = c("Softsat", "Indifferent", "Complex")),
    Mode = "Web"
  )


socdem_sensan <- c(
  "TGANC", "AGET1", "AGET3", "GENF", "MIGY", "EDTERT", "EDNA", "EMPP", "EMPNO", "EDNA",
    "EMPRET", "EMPNA", "KIDY", "KIDNA", "INCT1", "INCT3", "INCT3","INCNA", "PCONS"
)

w1_sensan <- rbind(wave1_cawi_sensan |> select(pid, hid, Mode, LC, starts_with("cprob_"),
  all_of(socdem_sensan)),
             wave1_papi_sensan |> select(pid, hid, Mode, LC, starts_with("cprob_"),
  all_of(socdem_sensan)))

w1_sensan |> ai()
```

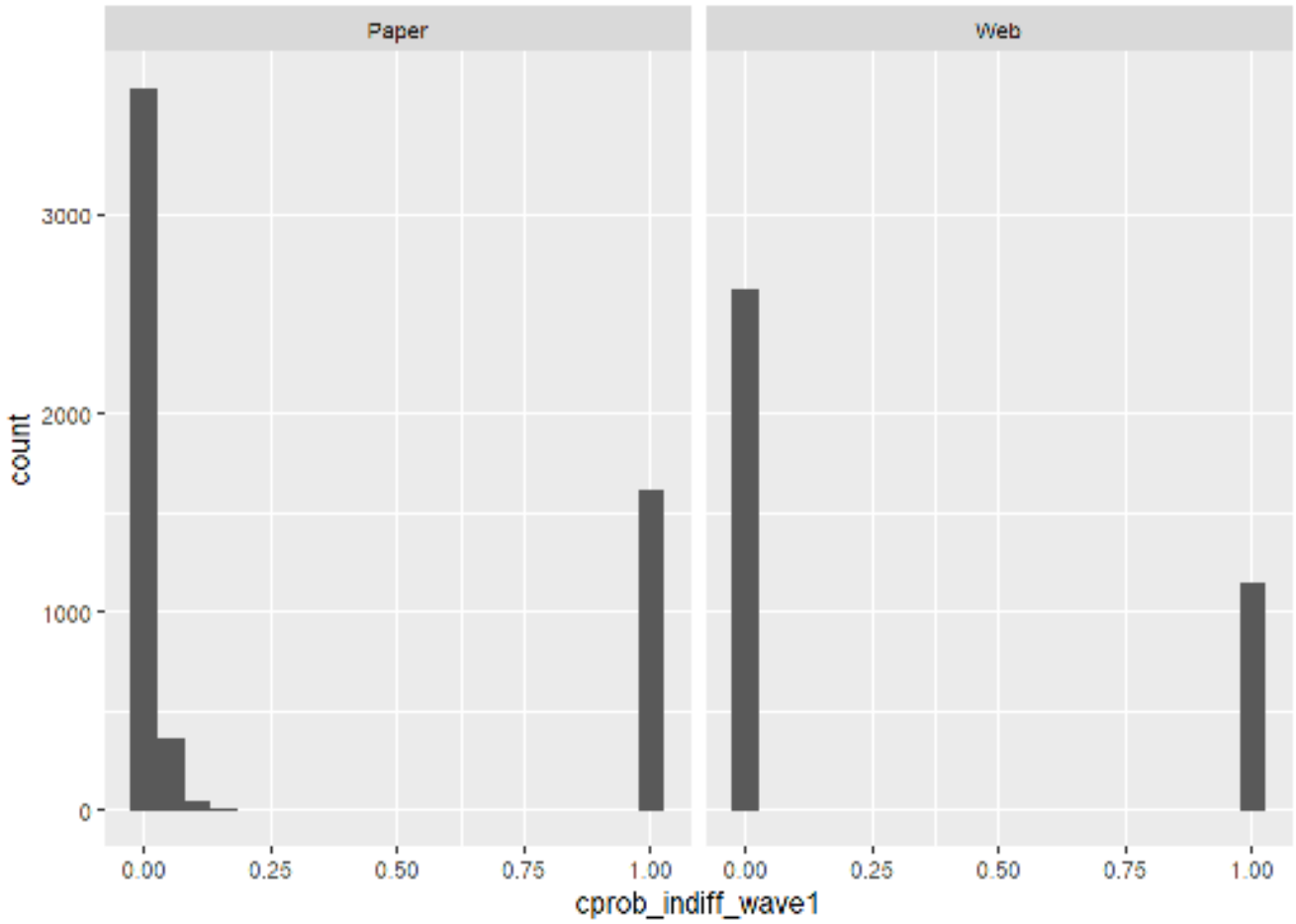
```
Rows: 6
Columns: 22
$ Mode         <chr> "Web", "Web", "Web", "Web", "Web", "Web"
$ LC           <fct> Softsat, Softsat, Indifferent, Softsat, Softsat, Softsat
$ cprob_indiff <dbl> 0.000, 0.000, 0.978, 0.000, 0.000, 0.000
$ cprob_opt    <dbl> 1.000, 1.000, 0.000, 1.000, 0.834, 1.000
$ cprob_3      <dbl> 0.000, 0.000, 0.022, 0.000, 0.166, 0.000
$ TGANC        <dbl> 2, 2, 2, 2, 2, 2
$ AGET1        <dbl> 1, 1, 2, 1, 2, 2
$ AGET3        <dbl> 1, 1, 1, 1, 1, 1
$ GENF         <dbl> 1, 2, 1, 1, 1, 1
$ MIGY         <dbl> 2, 1, 1, 2, 1, 1
$ EDTERT       <dbl> 2, 2, 1, 1, 2, 1
$ EDNA         <dbl> 1, 1, 1, 1, 1, 1
```

```
$ EMPP          <dbl> 1, 1, 1, 2, 1, 1
$ EMPNO         <dbl> 1, 1, 1, 1, 1, 1
$ EMPRET        <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA         <dbl> 1, 1, 1, 1, 1, 1
$ KIDY          <dbl> 2, 1, 2, 1, 2, 1
$ KIDNA         <dbl> 1, 1, 1, 1, 1, 1
$ INCT1         <dbl> 2, 1, 1, 2, 1, 2
$ INCT3         <dbl> 1, 2, 2, 1, 2, 1
$ INCNA         <dbl> 1, 1, 1, 1, 1, 1
$ PCONS         <dbl> 2, 2, 2, 2, 2, 2
```

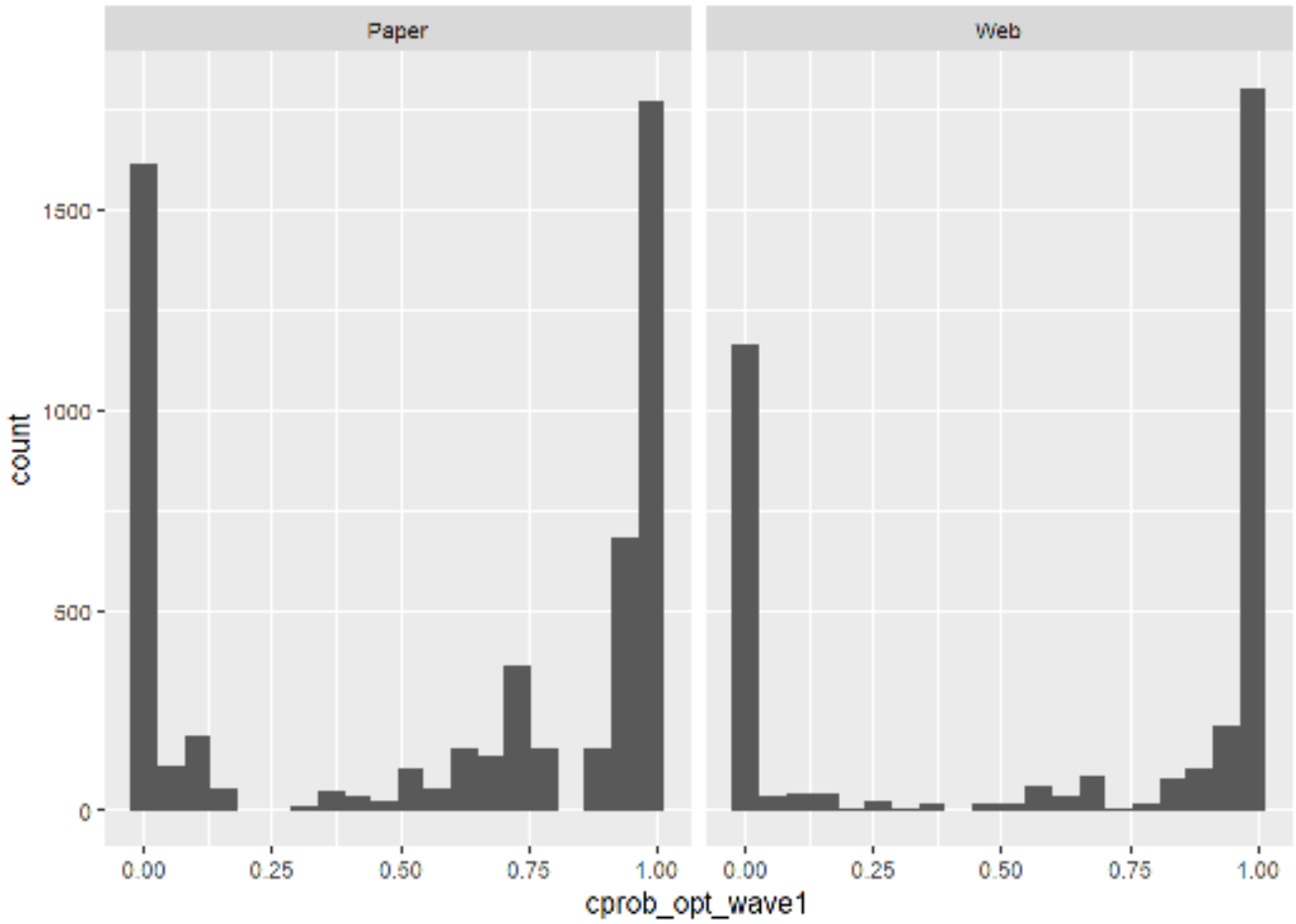
2 Distributions of Posterior Probs

- predicting the current cprob does not work, because the probabilities for Indiff and partly also Opti are factisch annähernd dichotom sind
- Wave 1:

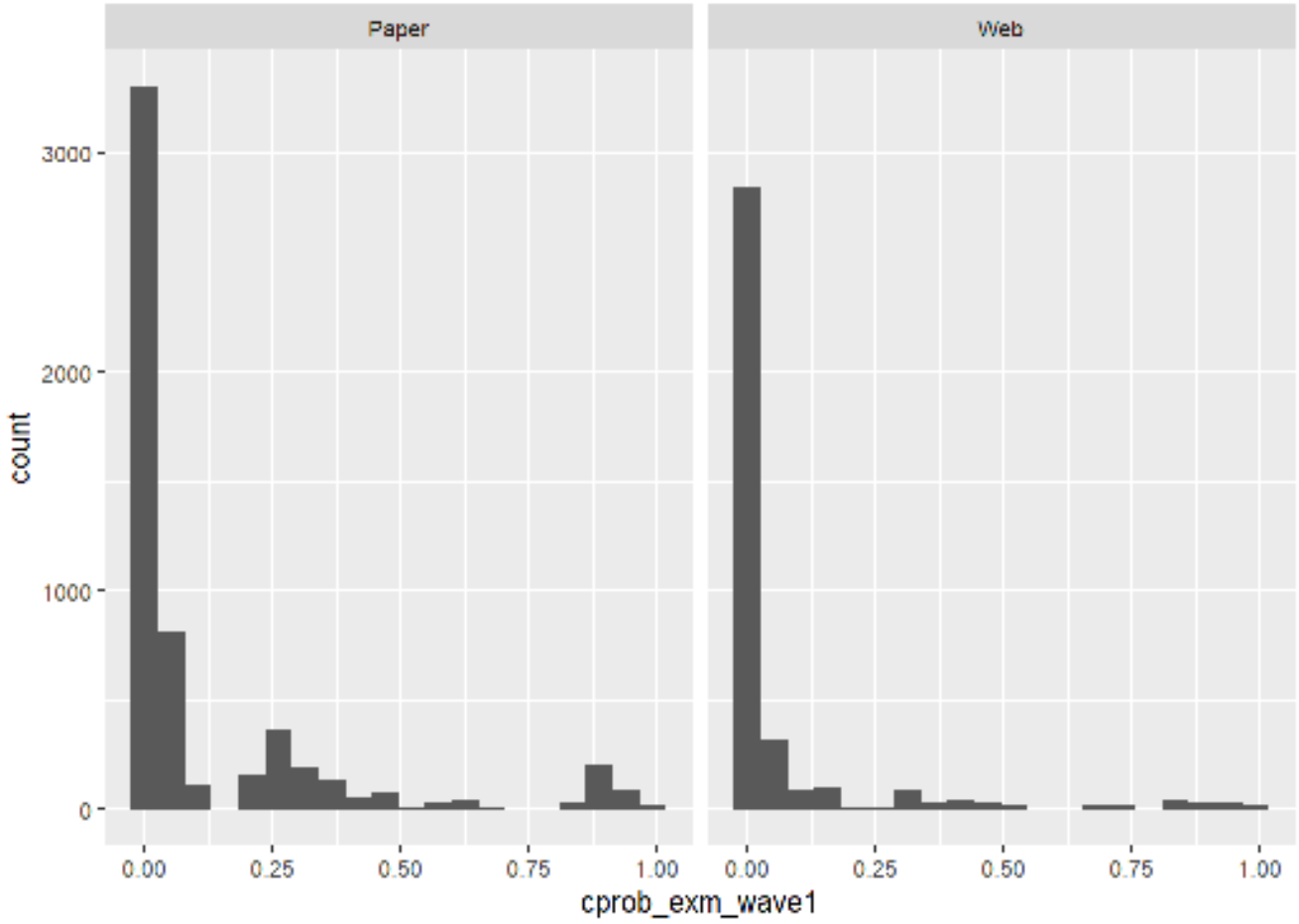
```
# distribution of cprob_indiff in wave1
df |>
  select(cprob_indiff_wave1, Mode_wave1) |>
  ggplot(aes(x = cprob_indiff_wave1)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave1)
```



```r
# for comparison: cprob_opt
df |>
  select(cprob_opt_wave1, Mode_wave1) |>
  ggplot(aes(x = cprob_opt_wave1)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave1)
```

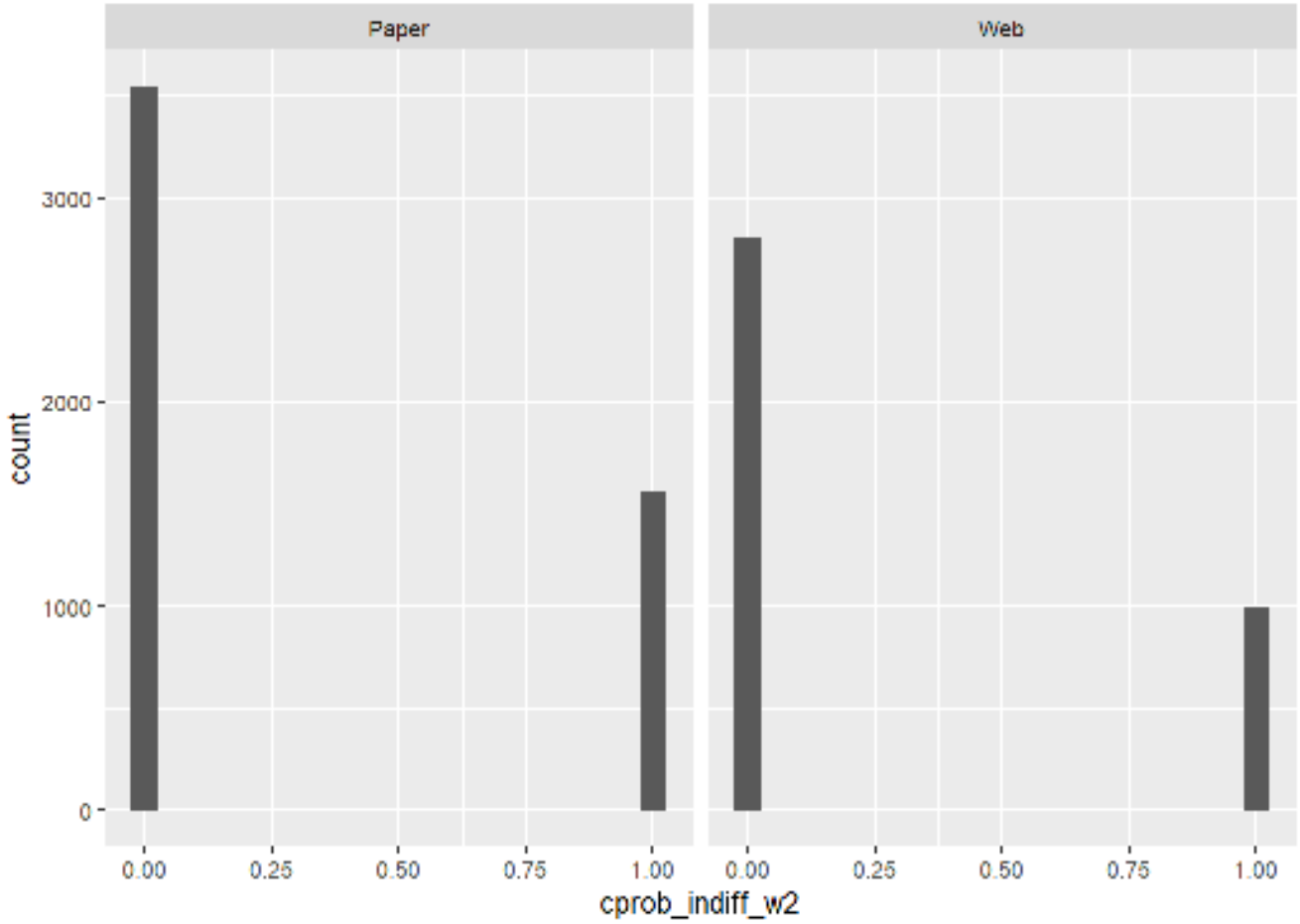



```r
# for comparison: cprob_exm
df |>
  select(cprob_exm_wave1, Mode_wave1) |>
  ggplot(aes(x = cprob_exm_wave1)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave1)
```

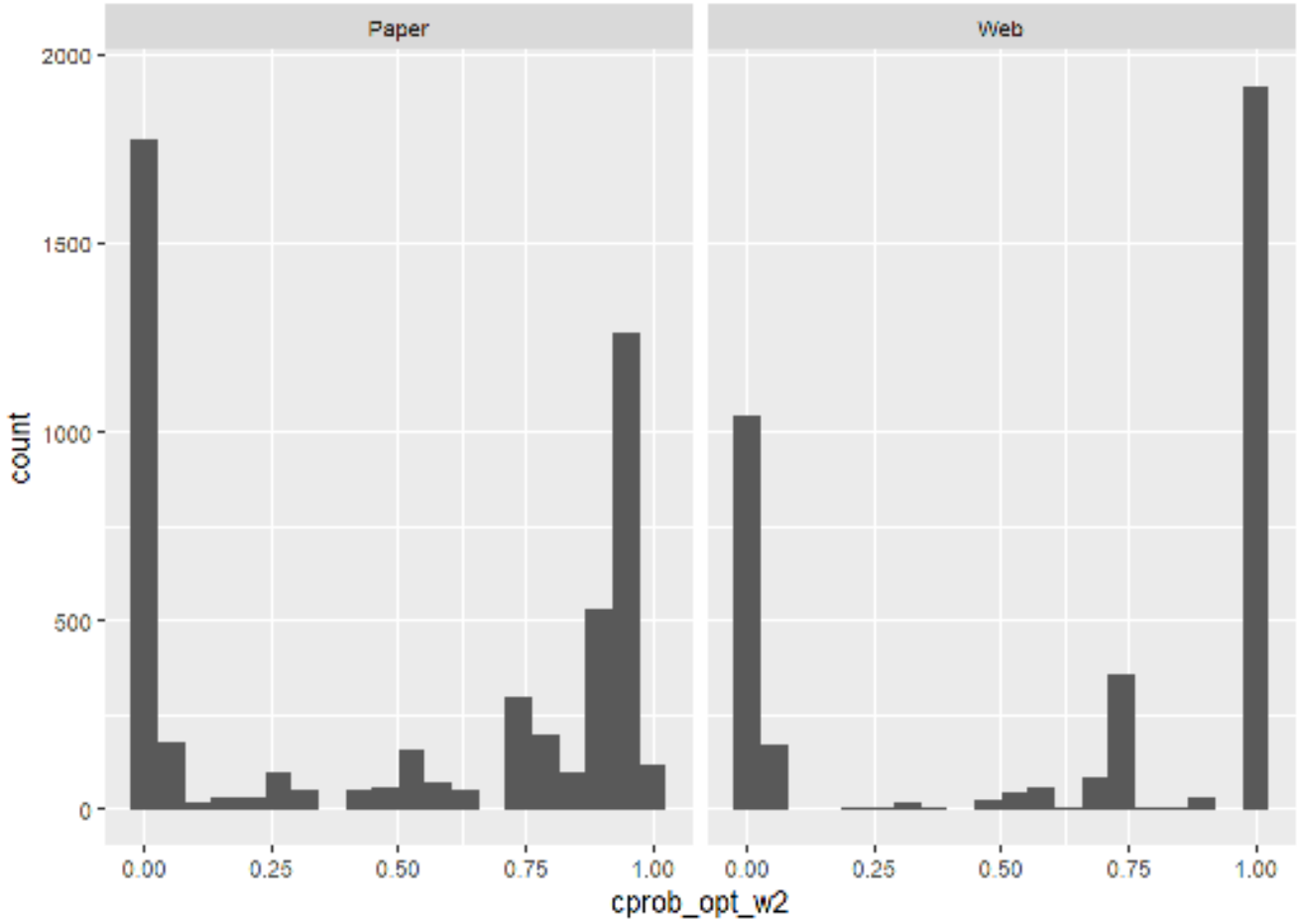


- W2:

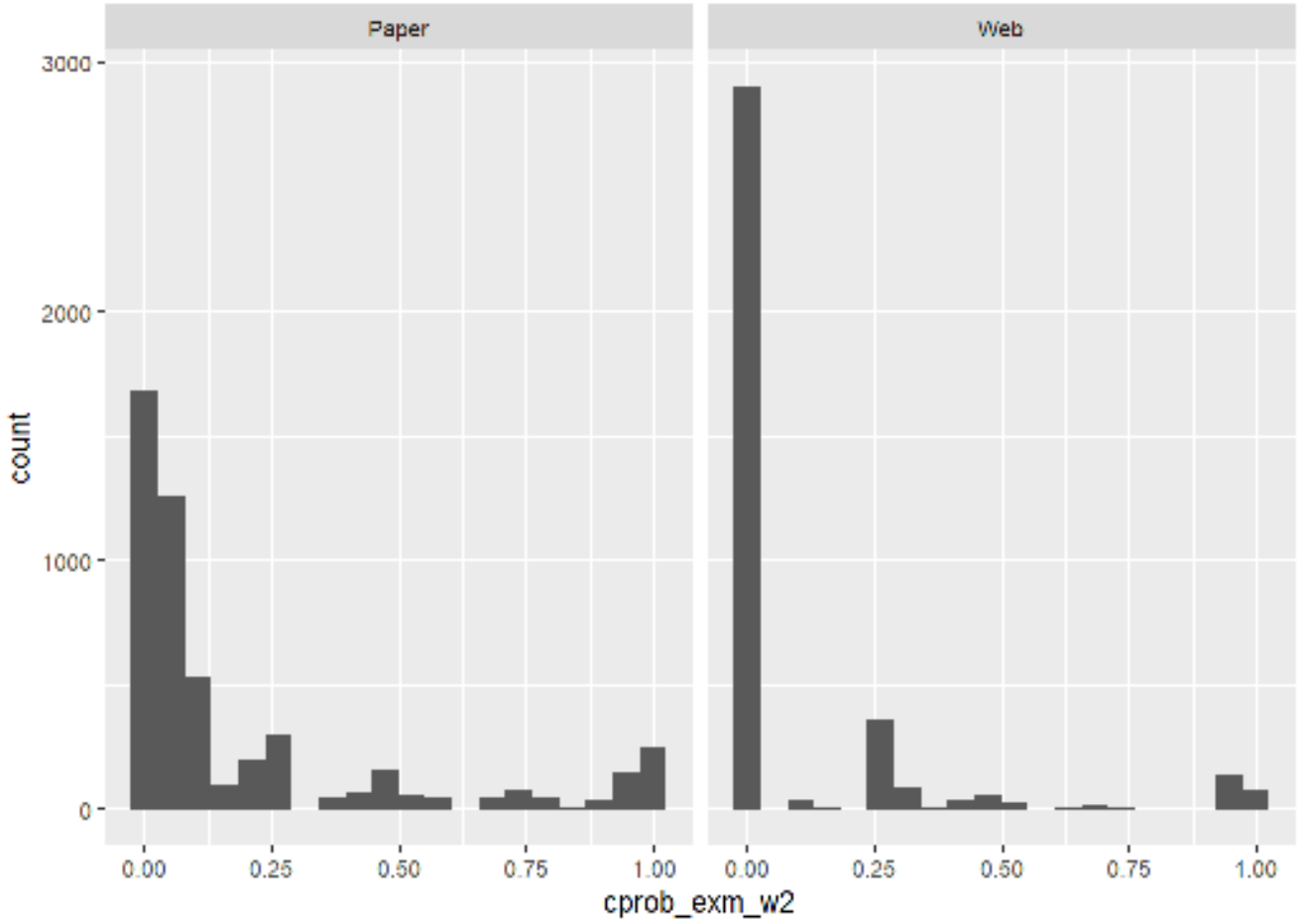
```
# distribution of cprob_indiff in w2
df |>
  filter(!is.na(Mode_w2)) |>
  select(cprob_indiff_w2, Mode_w2) |>
  ggplot(aes(x = cprob_indiff_w2)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_w2)
```




```
# for comparison: cprob_opt
df |>
  filter(!is.na(Mode_w2)) |>
  select(cprob_opt_w2, Mode_w2) |>
  ggplot(aes(x = cprob_opt_w2)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_w2)
```

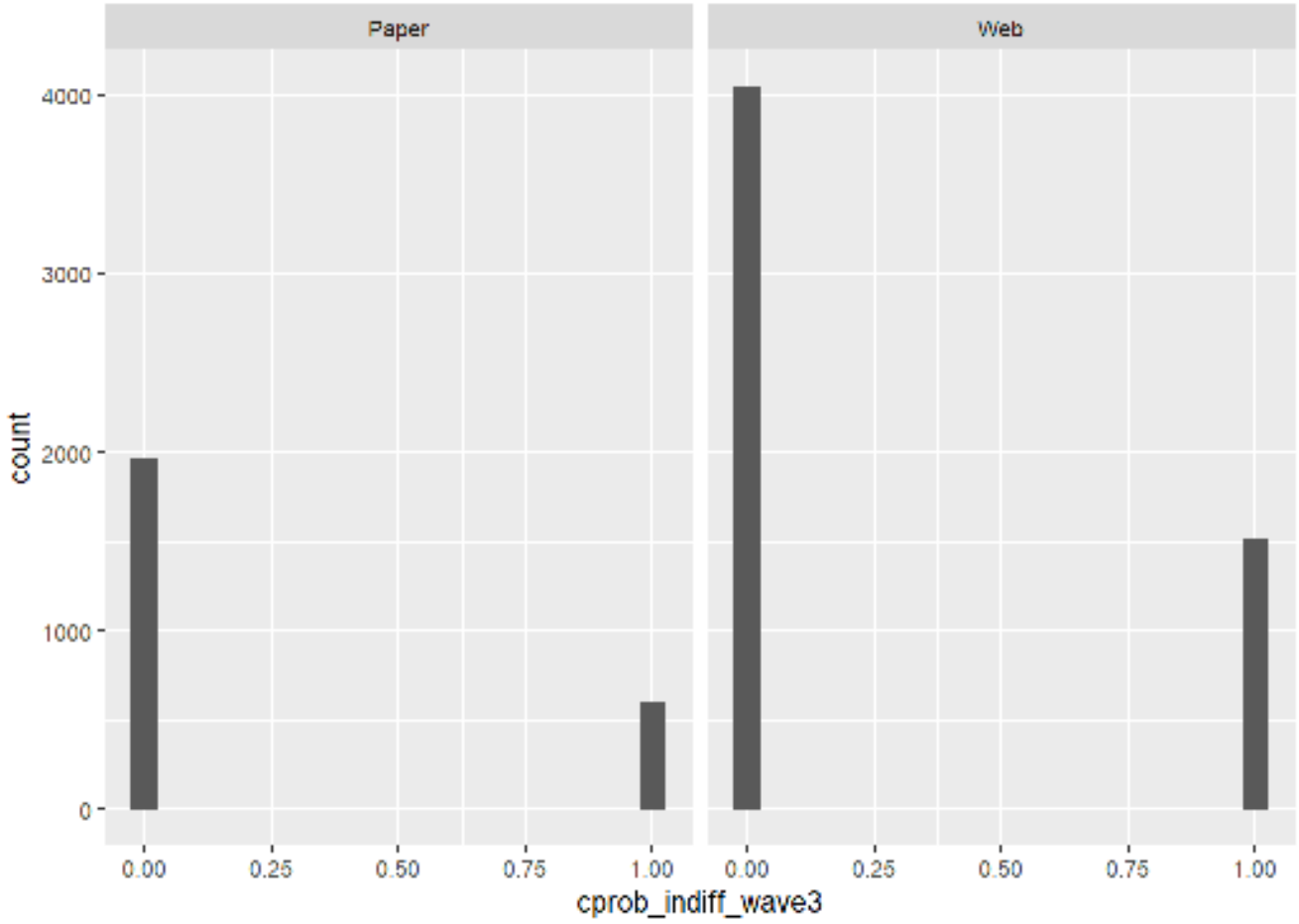



```
# for comparison: cprob_exm
df |>
  filter(!is.na(Mode_w2)) |>
  select(cprob_exm_w2, Mode_w2) |>
  ggplot(aes(x = cprob_exm_w2)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_w2)
```

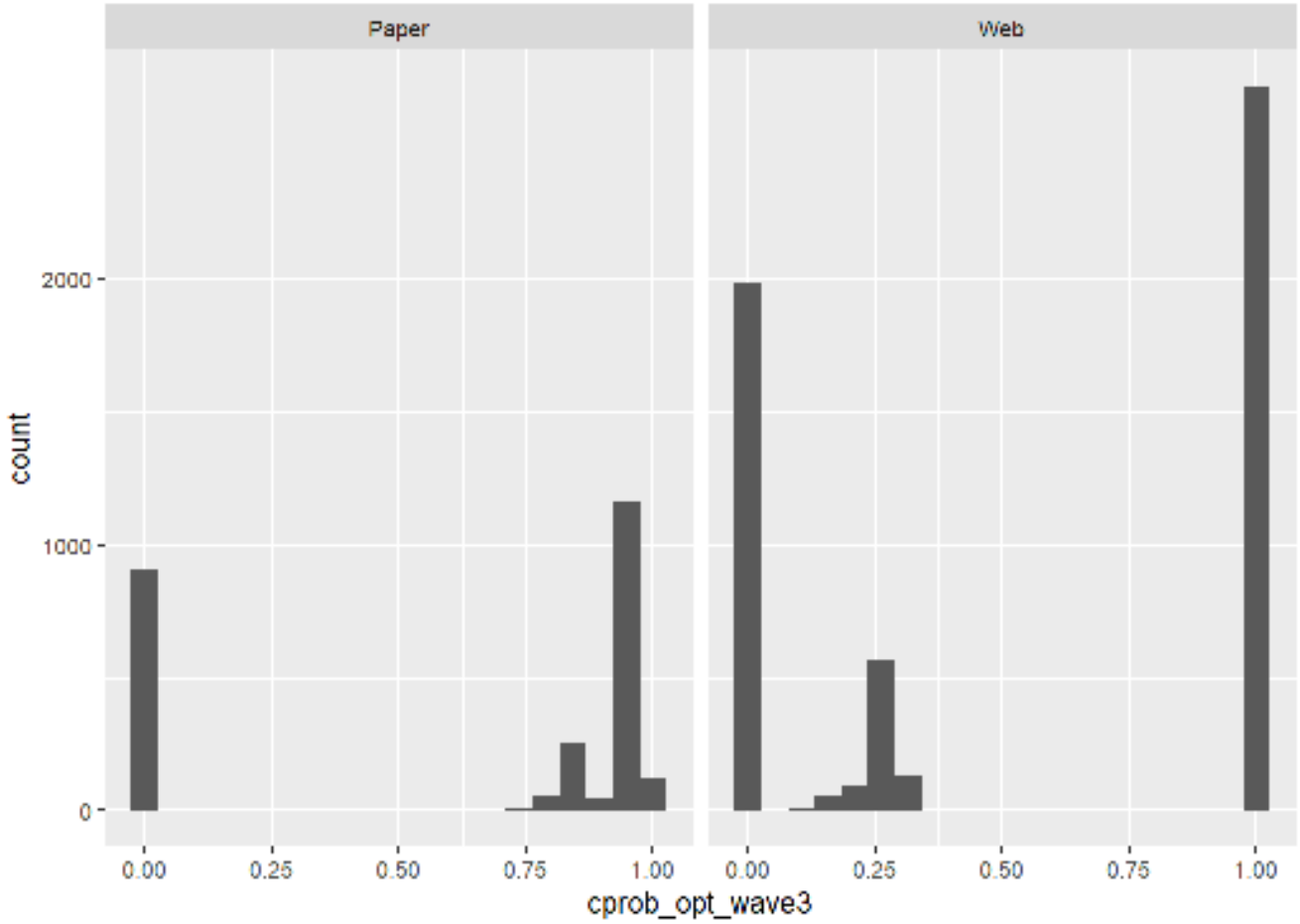


- Wave 3:

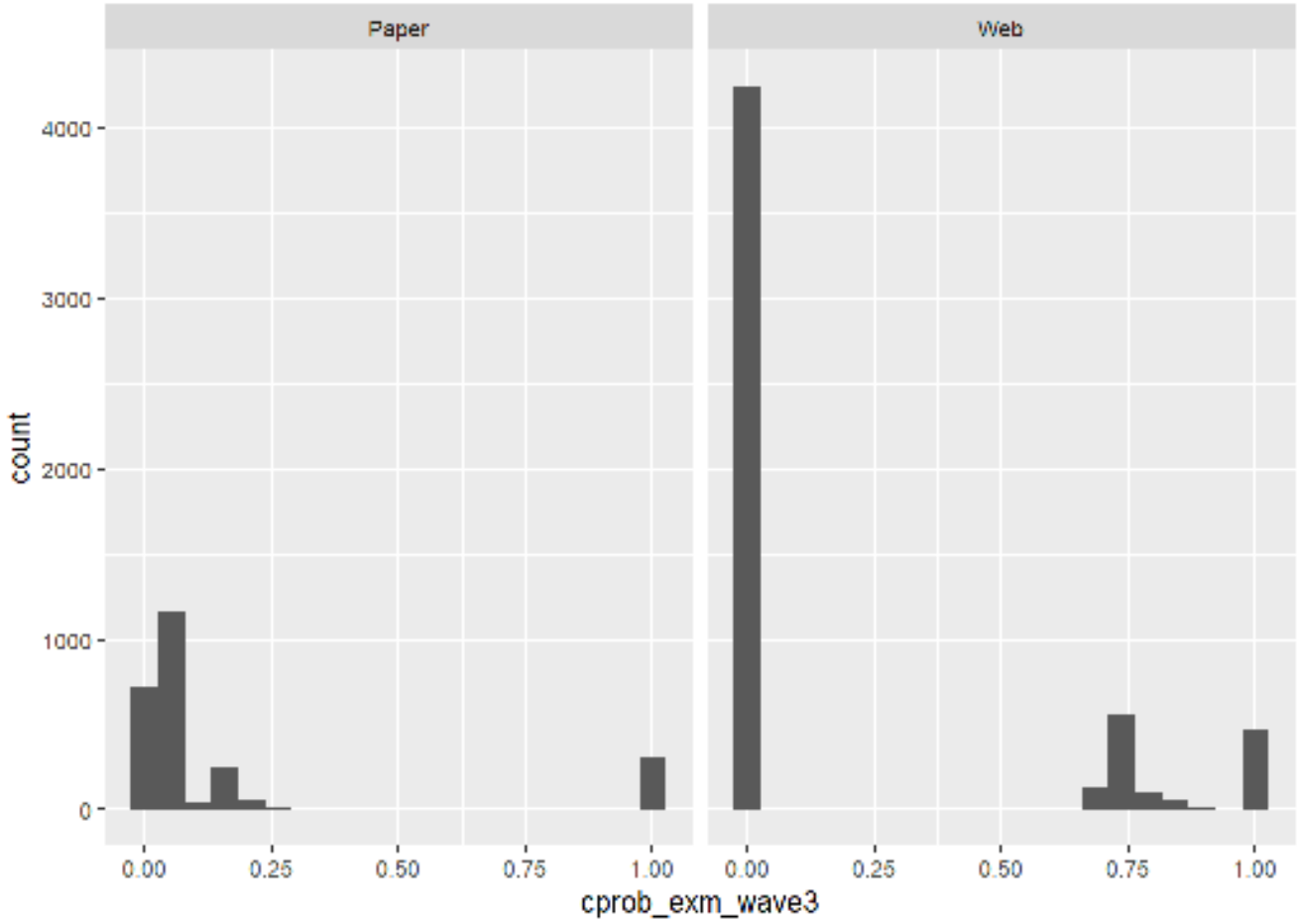
```
# distribution of cprob_indiff in wave3
df |>
  filter(!is.na(Mode_wave3)) |>
  select(cprob_indiff_wave3, Mode_wave3) |>
  ggplot(aes(x = cprob_indiff_wave3)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave3)
```




```
# for comparison: cprob_opt
df |>
  filter(!is.na(Mode_wave3)) |>
  select(cprob_opt_wave3, Mode_wave3) |>
  ggplot(aes(x = cprob_opt_wave3)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave3)
```




```
# for comparison: cprob_exm
df |>
  filter(!is.na(Mode_wave3)) |>
  select(cprob_exm_wave3, Mode_wave3) |>
  ggplot(aes(x = cprob_exm_wave3)) +
  geom_histogram(bins = 20) +
  facet_wrap(~ Mode_wave3)
```



2.0.1 Prediction: Which Model to Choose

```
# Given a model, predict values of yvar from xvar
# This function supports one predictor and one predicted variable
# xrange: If NULL, determine the x range from the model object. If a vector with
#   two numbers, use those as the min and max of the prediction range.
# samples: Number of samples across the x range.
# ...: Further arguments to be passed to predict()
predictvals <- function(model, xvar, yvar, xrange = NULL, samples = 100, ...) {

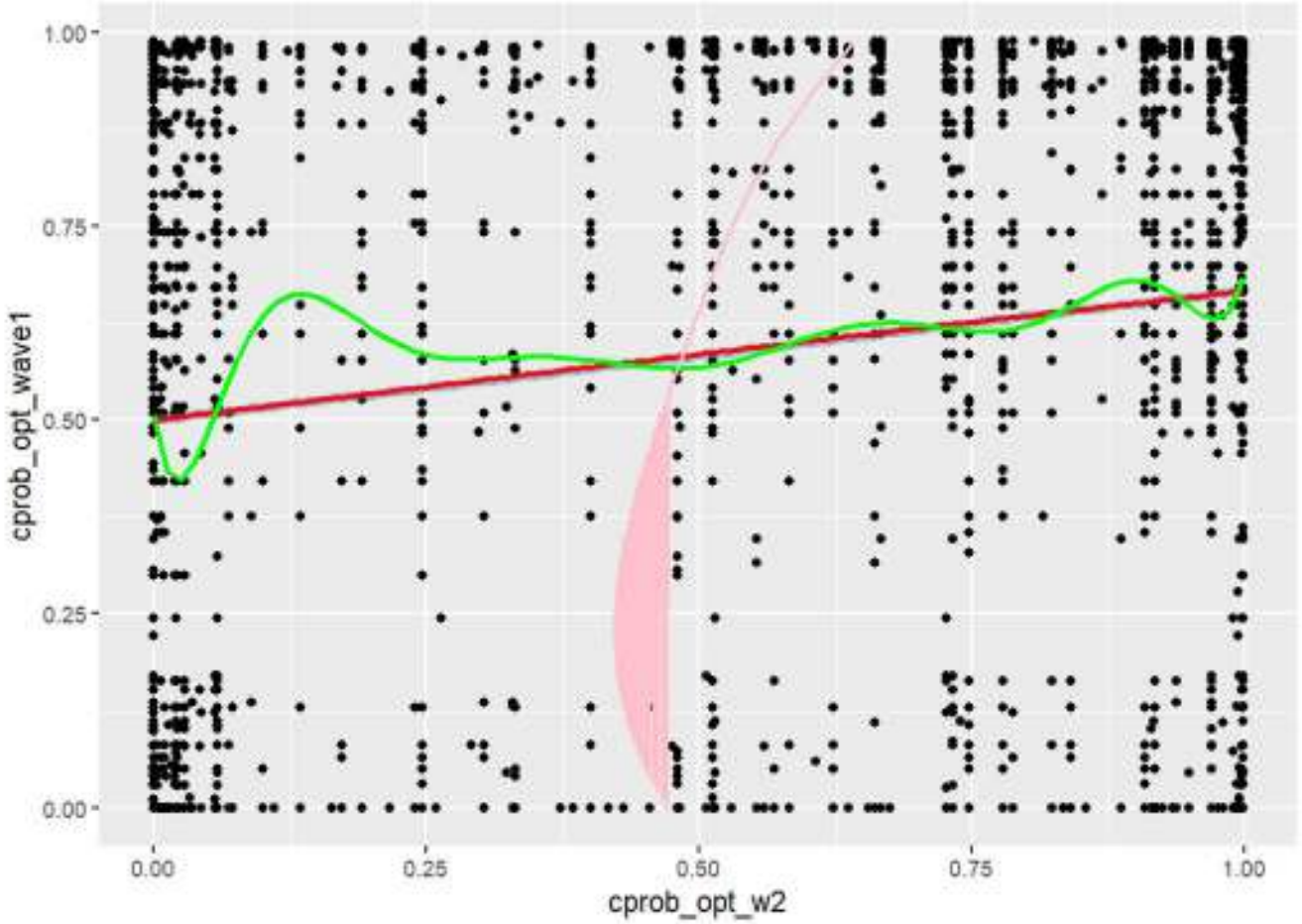
  # If xrange isn't passed in, determine xrange from the models.
  # Different ways of extracting the x range, depending on model type
  if (is.null(xrange)) {
    if (any(class(model) %in% c("lm", "glm")))
      xrange <- range(model$model[[xvar]])
    else if (any(class(model) %in% "loess"))
      xrange <- range(model$x)
  }

  newdata <- data.frame(x = seq(xrange[1], xrange[2], length.out = samples))
  names(newdata) <- xvar
  newdata[[yvar]] <- predict(model, newdata = newdata, ...)
  newdata
}
```

```r
geom_lm <- function(formula = y ~ x, colour = alpha("steelblue", 0.5),
                    linewidth = 2, ...) {
  geom_smooth(formula = formula, se = FALSE, method = "lm", colour = colour,
    linewidth = linewidth, ...)
}

mod_loes <- loess(cprob_opt_w2 ~ cprob_opt_wave1, df)
loess_predicted <- predictvals(mod_loes, "cprob_opt_wave1", "cprob_opt_w2")

#zshg_opti <-
  ggplot(df, aes(cprob_opt_w2, cprob_opt_wave1)) +
  geom_point() +
  geom_lm() +
  geom_lm(y ~ poly(x, 2), linewidth = 1, colour = "red") +
  geom_lm(y ~ poly(x, 10), linewidth = 1, colour = "green") +
  geom_line(data = loess_predicted, color = "pink", linewidth = .8)
```




```r
# ggsave(file =
#   "[redacted]",
#   zshg_opti,
#         width = 6,
#         height = 4,
#         dpi = 300)
```

3 Predicting Satisficing

```
p_load(modelsummary, rcompanion)


#   1. Dummy coding and factor levels
df <- df |>
  mutate(
    y_opt_w2    = as.integer(LC_w2 == "Softsat"),
    y_exm_w2    = as.integer(LC_w2 == "Extremist"),
    y_indiff_w2 = as.integer(LC_w2 == "Indifferent"),
    y_opt_wave3      = as.integer(LC_wave3   == "Softsat"),
    y_exm_wave3      = as.integer(LC_wave3   == "Extremist"),
    y_indiff_wave3   = as.integer(LC_wave3   == "Indifferent"),
    Mode_wave1     = factor(Mode_wave1, levels = c("Paper", "Web")),
    Mode_w2     = factor(Mode_w2, levels = c("Paper", "Web")),
    Mode_wave3       = factor(Mode_wave3,   levels = c("Paper", "Web"))
  )

#   2. Consistent analysis subsets
df_wave1_w2 <- df |> filter(!is.na(cprob_opt_wave1) & !is.na(LC_w2))
df_w2_wave3   <- df |> filter(!is.na(cprob_opt_w2) & !is.na(LC_wave3))
df_wave1_wave3   <- df |> filter(!is.na(cprob_opt_wave1) & !is.na(LC_wave3))

#   3. Model estimation


# Softsat
#wave 2 by wave 1
w2_opt <- glm(y_opt_w2 ~ cprob_opt_wave1    * Mode_wave1, data = df_wave1_w2, family =
 ↪  binomial)
#wave 3 by wave 2
wave3_opt_2 <- glm(y_opt_wave3   ~ cprob_opt_w2    * Mode_w2, data = df_w2_wave3,   family =
 ↪  binomial)
#wave 3 by wave 1
wave3_opt_1 <- glm(y_opt_wave3   ~ cprob_opt_wave1    * Mode_wave1, data = df_wave1_wave3,
 ↪  family = binomial)

# Extremist
w2_exm <- glm(y_exm_w2 ~ cprob_exm_wave1   * Mode_wave1, data = df_wave1_w2, family = binomial)
wave3_exm_2 <- glm(y_exm_wave3   ~ cprob_exm_w2   * Mode_w2, data = df_w2_wave3,   family =
 ↪  binomial)
wave3_exm_1 <- glm(y_exm_wave3   ~ cprob_exm_wave1   * Mode_wave1, data = df_wave1_wave3,
 ↪  family = binomial)

# Indifferent
w2_ind <- glm(y_indiff_w2 ~ cprob_indiff_wave1 * Mode_wave1, data = df_wave1_w2, family =
 ↪  binomial)
wave3_ind_2 <- glm(y_indiff_wave3   ~ cprob_indiff_w2 * Mode_w2, data = df_w2_wave3,   family =
 ↪  binomial)
wave3_ind_1 <- glm(y_indiff_wave3   ~ cprob_indiff_wave1 * Mode_wave1, data = df_wave1_wave3,
 ↪  family = binomial)

#   4. Regression tables
fmt <- function(x) format(round(x, 3), big.mark = ",")
gm  <- list(list("raw" = "nobs", "clean" = "N", "fmt" = fmt))
```

```
modelsummary(
  list("W2 by Wave 1" = w2_opt, "Wave 3 by W2" = wave3_opt_2, "Wave 3 by Wave 1" =
   ↪  wave3_opt_1),
  title        = "Softsat",
  exponentiate = TRUE, vcov = "HC1", cluster = ~ hid,
  stars        = c("*" = .05, "**" = .01, "***" = .001),
  statistic    = "conf.int", gof_map = gm, output = "flextable"
)
```

	W2 by Wave 1	Wave 3 by W2	Wave 3 by Wave 1
(Intercept)	0.829***	0.897*	0.927
	[0.757, 0.907]	[0.818, 0.984]	[0.843, 1.020]
cprob_opt_wave1	2.089***		1.682***
	[1.839, 2.373]		[1.472, 1.923]
Mode_wave1Web	1.522***		0.815**
	[1.316, 1.761]		[0.701, 0.949]
cprob_opt_wave1 × Mode_wave1Web	0.971		1.025
	[0.795, 1.186]		[0.836, 1.258]
cprob_opt_w2		1.892***	
		[1.643, 2.179]	
Mode_w2Web		0.765***	
		[0.656, 0.891]	
cprob_opt_w2 × Mode_w2Web		1.073	
		[0.868, 1.327]	
N	8,900	7,580	8,127

* p < 0.05, ** p < 0.01, *** p < 0.001

```
modelsummary(
  list("W2 by Wave 1" = w2_exm, "Wave 3 by W2" = wave3_exm_2, "Wave 3 by Wave 1" =
   ↪  wave3_exm_1),
  title        = "Extremist",
  exponentiate = TRUE, vcov = "HC1", cluster = ~ hid,
  stars        = c("*" = .05, "**" = .01, "***" = .001),
```

```
  statistic    = "conf.int", gof_map = gm, output = "flextable"
)
```

	W2 by Wave 1	Wave 3 by W2	Wave 3 by Wave 1
(Intercept)	0.113***	0.173***	0.185***
	[0.102, 0.124]	[0.156, 0.190]	[0.170, 0.202]
cprob_exm_wave1	8.135***		3.830***
	[6.270, 10.555]		[2.928, 5.012]
Mode_wave1Web	0.598***		1.425***
	[0.504, 0.710]		[1.258, 1.614]
cprob_exm_wave1 × Mode_wave1Web	1.027		0.688
	[0.628, 1.680]		[0.436, 1.085]
cprob_exm_w2		3.000***	
		[2.366, 3.804]	
Mode_w2Web		1.532***	
		[1.340, 1.752]	
cprob_exm_w2 × Mode_w2Web		0.935	
		[0.640, 1.366]	
N	8,900	7,580	8,127

* p < 0.05, ** p < 0.01, *** p < 0.001

```
modelsummary(
  list("W2 by Wave 1" = w2_ind, "Wave 3 by W2" = wave3_ind_2, "Wave 3 by Wave 1" =
   ↪  wave3_ind_1),
  title        = "Indifferent",
  exponentiate = TRUE, vcov = "HC1", cluster = ~ hid,
  stars        = c("*" = .05, "**" = .01, "***" = .001),
  statistic    = "conf.int", gof_map = gm, output = "flextable"
)
```

	W2 by Wave 1	Wave 3 by W2	Wave 3 by Wave 1
(Intercept)	0.353***	0.321***	0.290***

	W2 by Wave 1	Wave 3 by W2	Wave 3 by Wave 1
	[0.328, 0.379]	[0.295, 0.349]	[0.268, 0.314]
cprob_indiff_wave1	2.019***		1.806***
	[1.692, 2.176]		[1.573, 2.072]
Mode_wave1Web	0.808***		0.965
	[0.717, 0.911]		[0.849, 1.097]
cprob_indiff_wave1 × Mode_wave1Web	1.004		1.129
	[0.821, 1.228]		[0.912, 1.397]
cprob_indiff_w2		1.500***	
		[1.301, 1.728]	
Mode_w2Web		0.900	
		[0.793, 1.023]	
cprob_indiff_w2 × Mode_w2Web		1.142	
		[0.914, 1.427]	
N	8,900	7,580	8,127

* p < 0.05, ** p < 0.01, *** p < 0.001

```
models_named <- list(
  "Opt:  W2 by Wave 1" = w2_opt, "Opt:  Wave 3 by W2" = wave3_opt_2, "Opt:  Wave 3 by Wave 1" =
   ↪  wave3_opt_1,
  "ExM:  W2 by Wave 1" = w2_exm, "ExM:  Wave 3 by W2" = wave3_exm_2, "ExM:  Wave 3 by Wave 1" =
   ↪  wave3_exm_1,
  "Ind:  W2 by Wave 1" = w2_ind, "Ind:  Wave 3 by W2" = wave3_ind_2, "Ind:  Wave 3 by Wave 1" =
   ↪  wave3_ind_1
)

imap(models_named, ~ {
  cat("\n ", .y, " \n")
  print(rcompanion::nagelkerke(.x)$Pseudo.R.squared.for.model.vs.null)
})
```

```
  Opt:  W2 by Wave 1
                         Pseudo.R.squared
McFadden                         0.0249764
Cox and Snell (ML)               0.0331143
```

```
Nagelkerke (Cragg and Uhler)        0.0447302

  Opt:  Wave 3 by W2
                            Pseudo.R.squared
McFadden                           0.0159518
Cox and Snell (ML)                 0.0217865
Nagelkerke (Cragg and Uhler)       0.0291015

  Opt:  Wave 3 by Wave 1
                            Pseudo.R.squared
McFadden                           0.0107928
Cox and Snell (ML)                 0.0147873
Nagelkerke (Cragg and Uhler)       0.0197557

  ExM:  W2 by Wave 1
                            Pseudo.R.squared
McFadden                           0.0599039
Cox and Snell (ML)                 0.0421532
Nagelkerke (Cragg and Uhler)       0.0822128

  ExM:  Wave 3 by W2
                            Pseudo.R.squared
McFadden                           0.0200450
Cox and Snell (ML)                 0.0199614
Nagelkerke (Cragg and Uhler)       0.0314707

  ExM:  Wave 3 by Wave 1
                            Pseudo.R.squared
McFadden                           0.0160436
Cox and Snell (ML)                 0.0159585
Nagelkerke (Cragg and Uhler)       0.0252062

  Ind:  W2 by Wave 1
                            Pseudo.R.squared
McFadden                           0.0172981
Cox and Snell (ML)                 0.0205100
Nagelkerke (Cragg and Uhler)       0.0293753

  Ind:  Wave 3 by W2
                            Pseudo.R.squared
McFadden                          0.00805979
Cox and Snell (ML)                0.00921731
Nagelkerke (Cragg and Uhler)      0.01349490

  Ind:  Wave 3 by Wave 1
                            Pseudo.R.squared
McFadden                           0.0152295
Cox and Snell (ML)                 0.0173262
Nagelkerke (Cragg and Uhler)       0.0253820

$`Opt:  W2 by Wave 1`
                            Pseudo.R.squared
McFadden                           0.0249764
Cox and Snell (ML)                 0.0331143
Nagelkerke (Cragg and Uhler)       0.0447302

$`Opt:  Wave 3 by W2`
```

```
                            Pseudo.R.squared
McFadden                          0.0159518
Cox and Snell (ML)                0.0217865
Nagelkerke (Cragg and Uhler)      0.0291015

$`Opt:  Wave 3 by Wave 1`
                            Pseudo.R.squared
McFadden                          0.0107928
Cox and Snell (ML)                0.0147873
Nagelkerke (Cragg and Uhler)      0.0197557

$`ExM:  W2 by Wave 1`
                            Pseudo.R.squared
McFadden                          0.0599039
Cox and Snell (ML)                0.0421532
Nagelkerke (Cragg and Uhler)      0.0822128

$`ExM:  Wave 3 by W2`
                            Pseudo.R.squared
McFadden                          0.0200450
Cox and Snell (ML)                0.0199614
Nagelkerke (Cragg and Uhler)      0.0314707

$`ExM:  Wave 3 by Wave 1`
                            Pseudo.R.squared
McFadden                          0.0160436
Cox and Snell (ML)                0.0159585
Nagelkerke (Cragg and Uhler)      0.0252062

$`Ind:  W2 by Wave 1`
                            Pseudo.R.squared
McFadden                          0.0172981
Cox and Snell (ML)                0.0205100
Nagelkerke (Cragg and Uhler)      0.0293753

$`Ind:  Wave 3 by W2`
                            Pseudo.R.squared
McFadden                         0.00805979
Cox and Snell (ML)               0.00921731
Nagelkerke (Cragg and Uhler)     0.01349490

$`Ind:  Wave 3 by Wave 1`
                            Pseudo.R.squared
McFadden                          0.0152295
Cox and Snell (ML)                0.0173262
Nagelkerke (Cragg and Uhler)      0.0253820
```

```
#   Opt:  W2 by Wave 1
```

```
# Purpose: build the LaTeX table of the logistic regressions directly from
# the fitted models; the table layout is described once and all numbers are
# taken from the model objects at render time.

#   Helper functions
nagR2 <- function(model) {
  round(rcompanion::nagelkerke(model)$Pseudo.R.squared.for.model.vs.null[
    "Nagelkerke (Cragg and Uhler)", "Pseudo.R.squared"], 2)
```

```
}

mf <- function(model) {
  round(rcompanion::nagelkerke(model)$Pseudo.R.squared.for.model.vs.null[
    "McFadden", "Pseudo.R.squared"], 2)
}

get_stars <- function(model, term_idx) {
  se <- sqrt(diag(sandwich::vcovCL(model, cluster = ~ hid)))[term_idx]
  z  <- abs(coef(model)[term_idx] / se)
  p  <- 2 * pnorm(z, lower.tail = FALSE)
  case_when(p < .001 ~ "***", p < .01 ~ "**", p < .05 ~ "*", TRUE ~ "")
}

extract <- function(model) {
  map(seq_along(coef(model)), ~ list(
    or    = round(exp(coef(model)[.x]), 3),
    ci_lo = round(exp(confint.default(model)[.x, 1]), 3),
    ci_hi = round(exp(confint.default(model)[.x, 2]), 3),
    stars = get_stars(model, .x)
  ))
}

num <- function(x) paste0("\\num{", x, "}")

#   Which models form the columns / blocks, and how they are labelled
tab_spec <- list(
  caption = paste("Logistic Regression Results of Most Probable Satisficing",
                  "Pattern by Previous Likelihood of Same Satisficing"),
  label   = "tab:PredictionRegression",
  colheads = c("Logistic Parameter$^a$",
               "Wave 2 by 1.1", "Wave 2 by 2", "Wave 3 by 1.1"),
  firstcol = "OR [95\\% CI]",
  blocks = list(
    list(pattern = "Softsat",
         models  = list(w2_opt, wave3_opt_2, wave3_opt_1),
         terms   = c("(Intercept)",
                     "$\\hat{p}_{Softsat}$",
                     "Web Mode (Ref: Paper)",
                     "$\\hat{p}_{Softsat} \\times$ Web Mode")),
    list(pattern = "Extremist",
         models  = list(w2_exm, wave3_exm_2, wave3_exm_1),
         terms   = c("(Intercept)",
                     "$\\hat{p}_{Extremist}$",
                     "Web Mode (Ref: Paper)",
                     "$\\hat{p}_{Extremist} \\times$ Web Mode")),
    list(pattern = "Indifferents",
         models  = list(w2_ind, wave3_ind_2, wave3_ind_1),
         terms   = c("(Intercept)",
                     "$\\hat{p}_{Indifferents}$",
                     "Web Mode (Ref: Paper)",
                     "$\\hat{p}_{Indifferents} \\times$ Web Mode"))
  ),
  note = paste("OR = Odds Ratio. CI = confidence interval",
               "[lower level, upper level]. $\\hat{p}$ = model estimated posterior",
               "class probability of previous wave. $\\hat{c}_{modal}$ = most likely class",
```

```
              "membership. Robust standard errors with individuals clustered in",
              "households (HC1).")
)

#   Row builders
or_row <- function(ex, i) {
  paste0(" & ", paste(map_chr(ex, ~ paste0(num(.x[[i]]$or), .x[[i]]$stars)), collapse = " & "),
   ↪  " \\\\")
}

ci_row <- function(ex, i) {
  paste0(" & ", paste(map_chr(ex, ~ paste0("{}[", num(.x[[i]]$ci_lo), ", ",
                                           num(.x[[i]]$ci_hi), "]")), collapse = " & "), "
                                            ↪  \\\\")
}

fit_row <- function(models, fun) {
  paste0(" & ", paste(map_chr(models, ~ num(fun(.x))), collapse = " & "), " \\\\")
}

block_rows <- function(block) {
  ex <- map(block$models, extract)
  out <- c(paste0(" & \\multicolumn{3}{c}{$\\hat{c}_{modal} =$ ", block$pattern, "} \\\\"))
  for (i in seq_along(block$terms)) {
    out <- c(out,
             paste0(block$terms[i], or_row(ex, i)),
             ci_row(ex, i))
  }
  out <- c(out,
           paste0("N", fit_row(block$models, function(m) nrow(m$model))),
           paste0("McFadden $R^2$", fit_row(block$models, mf)),
           paste0("Nagelkerke $R^2$", fit_row(block$models, nagR2)))
  out
}

#   Assemble the LaTeX table
latex <- c(
  "\\begin{table}",
  "  \\begin{threeparttable}",
  paste0("  \\caption{", tab_spec$caption, "}"),
  paste0("  \\label{", tab_spec$label, "}"),
  "  \\begin{tabular}[t]{@{}lccc@{}}",
  "  \\toprule",
  paste0("  ", paste(tab_spec$colheads, collapse = " & "), " \\\\ \\cline{2-4}"),
  paste0("  & ", paste(rep(tab_spec$firstcol, 3), collapse = " & "), " \\\\ \\midrule"),
  unlist(map(tab_spec$blocks, block_rows)),
  "  \\end{tabular}",
  "  \\begin{tablenotes}[flushleft]",
  paste0("    \\item \\footnotesize \\textit{Note.} ", tab_spec$note),
  "    \\textsuperscript{a} Parameters refer to the characteristics of previous waves.",
  "    \\textsuperscript{*}p$<0.05$, \\textsuperscript{**}p$<0.01$,
   ↪  \\textsuperscript{***}p$<0.001$.",
  "  \\end{tablenotes}",
  "  \\end{threeparttable}",
  "\\end{table}"
)
```

```
# write out for the manuscript, and show it in the rendered document
#writeLines(latex, "table1_satisficing.tex")

print(latex)
```

```
 [1] "\\begin{table}"
 [2] "  \\begin{threeparttable}"
 [3] "  \\caption{Logistic Regression Results of Most Probable Satisficing Pattern by Previous Likelihood
 [4] "  \\label{tab:PredictionRegression}"
 [5] "  \\begin{tabular}[t]{@{}lccc@{}}"
 [6] "  \\toprule"
 [7] "  Logistic Parameter$^a$ & Wave 2 by 1.1 & Wave 2 by 2 & Wave 3 by 1.1 \\\\ \\cline{2-4}"
 [8] "  & OR [95\\% CI] & OR [95\\% CI] & OR [95\\% CI] \\\\ \\midrule"
 [9] " & \\multicolumn{3}{c}{$\\hat{c}_{modal} =$ Softsat} \\\\"
[10] "(Intercept) & \\num{0.829}*** & \\num{0.897}* & \\num{0.927} \\\\"
[11] " & {}[\\num{0.757}, \\num{0.907}] & {}[\\num{0.818}, \\num{0.983}] & {}[\\num{0.843}, \\num{1.019}]
[12] "$\\hat{p}_{Softsat}$ & \\num{2.089}*** & \\num{1.892}*** & \\num{1.682}*** \\\\"
[13] " & {}[\\num{1.839}, \\num{2.373}] & {}[\\num{1.643}, \\num{2.179}] & {}[\\num{1.472}, \\num{1.923}]
[14] "Web Mode (Ref: Paper) & \\num{1.522}*** & \\num{0.765}*** & \\num{0.815}** \\\\"
[15] " & {}[\\num{1.316}, \\num{1.76}] & {}[\\num{0.657}, \\num{0.89}] & {}[\\num{0.701}, \\num{0.948}] \\
[16] "$\\hat{p}_{Softsat} \\times$ Web Mode & \\num{0.971} & \\num{1.073} & \\num{1.025} \\\\"
[17] " & {}[\\num{0.795}, \\num{1.187}] & {}[\\num{0.868}, \\num{1.326}] & {}[\\num{0.836}, \\num{1.257}]
[18] "N & \\num{8900} & \\num{7580} & \\num{8127} \\\\"
[19] "McFadden $R^2$ & \\num{0.02} & \\num{0.02} & \\num{0.01} \\\\"
[20] "Nagelkerke $R^2$ & \\num{0.04} & \\num{0.03} & \\num{0.02} \\\\"
[21] " & \\multicolumn{3}{c}{$\\hat{c}_{modal} =$ Extremist} \\\\"
[22] "(Intercept) & \\num{0.113}*** & \\num{0.173}*** & \\num{0.185}*** \\\\"
[23] " & {}[\\num{0.102}, \\num{0.124}] & {}[\\num{0.156}, \\num{0.19}] & {}[\\num{0.17}, \\num{0.202}] \\
[24] "$\\hat{p}_{Extremist}$ & \\num{8.135}*** & \\num{3}*** & \\num{3.83}*** \\\\"
[25] " & {}[\\num{6.265}, \\num{10.564}] & {}[\\num{2.374}, \\num{3.791}] & {}[\\num{2.926}, \\num{5.014}]
[26] "Web Mode (Ref: Paper) & \\num{0.598}*** & \\num{1.532}*** & \\num{1.425}*** \\\\"
[27] " & {}[\\num{0.503}, \\num{0.711}] & {}[\\num{1.34}, \\num{1.752}] & {}[\\num{1.257}, \\num{1.615}] \
[28] "$\\hat{p}_{Extremist} \\times$ Web Mode & \\num{1.027} & \\num{0.935} & \\num{0.688} \\\\"
[29] " & {}[\\num{0.622}, \\num{1.695}] & {}[\\num{0.64}, \\num{1.368}] & {}[\\num{0.434}, \\num{1.092}] \
[30] "N & \\num{8900} & \\num{7580} & \\num{8127} \\\\"
[31] "McFadden $R^2$ & \\num{0.06} & \\num{0.02} & \\num{0.02} \\\\"
[32] "Nagelkerke $R^2$ & \\num{0.08} & \\num{0.03} & \\num{0.03} \\\\"
[33] " & \\multicolumn{3}{c}{$\\hat{c}_{modal} =$ Indifferents} \\\\"
[34] "(Intercept) & \\num{0.353}*** & \\num{0.321}*** & \\num{0.29}*** \\\\"
[35] " & {}[\\num{0.328}, \\num{0.379}] & {}[\\num{0.295}, \\num{0.349}] & {}[\\num{0.268}, \\num{0.314}]
[36] "$\\hat{p}_{Indifferents}$ & \\num{1.919}*** & \\num{1.5}*** & \\num{1.806}*** \\\\"
[37] " & {}[\\num{1.692}, \\num{2.176}] & {}[\\num{1.301}, \\num{1.728}] & {}[\\num{1.574}, \\num{2.072}]
[38] "Web Mode (Ref: Paper) & \\num{0.808}*** & \\num{0.9} & \\num{0.965} \\\\"
[39] " & {}[\\num{0.717}, \\num{0.911}] & {}[\\num{0.793}, \\num{1.023}] & {}[\\num{0.849}, \\num{1.097}]
[40] "$\\hat{p}_{Indifferents} \\times$ Web Mode & \\num{1.004} & \\num{1.142} & \\num{1.129} \\\\"
[41] " & {}[\\num{0.821}, \\num{1.228}] & {}[\\num{0.914}, \\num{1.427}] & {}[\\num{0.912}, \\num{1.397}]
[42] "N & \\num{8900} & \\num{7580} & \\num{8127} \\\\"
[43] "McFadden $R^2$ & \\num{0.02} & \\num{0.01} & \\num{0.02} \\\\"
[44] "Nagelkerke $R^2$ & \\num{0.03} & \\num{0.01} & \\num{0.03} \\\\"
[45] "  \\end{tabular}"
[46] "  \\begin{tablenotes}[flushleft]"
[47] "    \\item \\footnotesize \\textit{Note.} OR = Odds Ratio. CI = confidence interval [lower level, up
[48] "    \\textsuperscript{a} Parameters refer to the characteristics of previous waves."
[49] "    \\textsuperscript{*}p$<0.05$, \\textsuperscript{**}p$<0.01$, \\textsuperscript{***}p$<0.001$."
[50] "  \\end{tablenotes}"
```

```
[51] "  \\end{threeparttable}"
[52] "\\end{table}"
```

4 Correlations Plots

```
#   Long-Format aus df
wave1_long <- df |>
  select(pid, hid, Mode = Mode_wave1,
         pSoftsat    = cprob_opt_wave1,
         pExtremist    = cprob_exm_wave1,
         pIndifferent = cprob_indiff_wave1) |>
  pivot_longer(cols = c(pSoftsat, pExtremist, pIndifferent),
               names_to = "variable", values_to = "epp") |>
  mutate(Wave = "Wave 1")

w2_long <- df |>
  filter(!is.na(cprob_opt_w2)) |>
  select(pid, hid, Mode = Mode_w2,
         pSoftsat    = cprob_opt_w2,
         pExtremist    = cprob_exm_w2,
         pIndifferent = cprob_indiff_w2) |>
  pivot_longer(cols = c(pSoftsat, pExtremist, pIndifferent),
               names_to = "variable", values_to = "epp") |>
  mutate(Wave = "Wave 2")

wave3_long <- df |>
  filter(!is.na(cprob_opt_wave3)) |>
  select(pid, hid, Mode = Mode_wave3,
         pSoftsat    = cprob_opt_wave3,
         pExtremist    = cprob_exm_wave3,
         pIndifferent = cprob_indiff_wave3) |>
  pivot_longer(cols = c(pSoftsat, pExtremist, pIndifferent),
               names_to = "variable", values_to = "epp") |>
  mutate(Wave = "Wave 3")
#   Plot function
make_cor_plot <- function(long_a, long_b, wave_a_label, wave_b_label, filename) {

  plot_df <- long_a |>
    select(pid, PatternA = variable, eppA = epp, ModeA = Mode) |>
    left_join(
      long_b |> select(pid, PatternB = variable, eppB = epp, ModeB = Mode),
      by = "pid",
      relationship = "many-to-many"
    ) |>
    filter(PatternA %in% c("pSoftsat", "pExtremist", "pIndifferent"),
           PatternB %in% c("pSoftsat", "pExtremist", "pIndifferent")) |>
    mutate(
      ModeA = ifelse(ModeA == "Web", paste(wave_a_label, "Web"), paste(wave_a_label, "Paper")),
      ModeB = ifelse(ModeB == "Web", paste(wave_b_label, "Web"), paste(wave_b_label, "Paper"))
    )

  p <- ggplot(plot_df, aes(x = eppA, y = eppB)) +
    geom_point(alpha = 0.1) +
    geom_smooth(formula = y ~ x, se = TRUE, method = "lm", color = "#1C7984", linewidth = 2) +
    stat_cor(p.accuracy = 0.001, method = "spearman",
```

```
            label.x = 0.02, label.y = -0.25,
            cor.coef.name = "rho", color = "black", size = 5.4, r.accuracy = 0.01) +
    labs(x = paste0("Model-estimated posterior probabilities at ", wave_a_label),
         y = paste0("Model-estimated posterior probabilities at ", wave_b_label)) +
    scale_x_continuous(breaks = seq(0, 1, by = 0.2)) +
    scale_y_continuous(breaks = seq(0, 1, by = 0.2), expand = c(0.1, 0.1)) +
    facet_grid(
      factor(ModeB, levels = c(paste(wave_b_label, "Web"), paste(wave_b_label, "Paper"))) +
      factor(PatternB, levels = c("pIndifferent", "pExtremist", "pSoftsat")) ~
      ModeA +
      factor(PatternA, levels = c("pSoftsat", "pExtremist", "pIndifferent"))
    ) +
    papaja::theme_apa() +
    theme(
      axis.title   = element_text(size = 17),
      axis.text.y  = element_text(size = 14, color = "black"),
      axis.text.x  = element_text(angle = 40, size = 14),
      strip.text   = element_text(size = 16)
    )

#  ggsave(filename, p,
#       width = 16, height = 13, dpi = 300)
  p
}

#   Create the plots
cors_w2_by_wave1 <- make_cor_plot(wave1_long, w2_long, "Wave 1", "Wave 2",
 ↪  "LC_Cors_w2_by_w1_RR.png")
cors_wave3_by_w2   <- make_cor_plot(w2_long, wave3_long,   "Wave 2", "Wave 3",
 ↪  "LC_Cors_w3_by_w2_RR.png")
cors_wave3_by_wave1   <- make_cor_plot(wave1_long, wave3_long,   "Wave 1", "Wave 3",
 ↪  "LC_Cors_w3_by_w1_RR.png")
```

5 Barplot on Omegas

```
p_load("readxl")

omegas_df <- read_excel(paste0(userpath,
 ↪  "[redacted]/Sample_Summaries.xlsx"),
     sheet = "Omegas") |>
  select(Wave:ProbOmega)


head(omegas_df,9)
```

```
# A tibble: 9 x 5
  Wave   Mode  LC        Indicator                    ProbOmega
  <chr>  <chr> <chr>     <chr>                            <dbl>
1 Wave 1 Web   Extremist Extremes selection               0.222
2 Wave 1 Web   Extremist Midpoint selection               0.282
3 Wave 1 Web   Extremist Open question nonresponse        0.5
4 Wave 1 Web   Extremist Closed question nonresponse      0.713
5 Wave 1 Web   Extremist Weak nondiff (often)             0.466
6 Wave 1 Web   Extremist Strong nondiff (sometimes)       0.77
```

```
7 Wave 1 Web   Extremist Strong nondiff (often)          0
8 Wave 1 Web   Extremist Break-off                      0.068
9 Wave 1 Web   Extremist Speeding                       0.08
```

```r
p_load(ggh4x)


n_labels <- bind_rows(
  df |> count(LC = LC_wave1, Mode = Mode_wave1) |> mutate(Wave = "Wave 1"),
  df |> count(LC = LC_w2, Mode = Mode_w2) |> mutate(Wave = "Wave 2"),
  df |> count(LC = LC_wave3,   Mode = Mode_wave3)   |> mutate(Wave = "Wave 3")
) |>
  filter(!is.na(LC), !is.na(Mode)) |>
  mutate(
    label = paste0("N = ", format(n, big.mark = ",", scientific = FALSE)),
    LC   = fct_relevel(LC,   "Softsat", "Indifferent", "Extremist"),
    Mode = fct_relevel(Mode, "Paper", "Web")
  )


cbPalette <- c("#000000", "#4F6E6A", "#CC79A7","#433C0A", "#39B4FF",
 ↪  "#F0E442","#D55E00","#61AD7B", "#7003A9" )


omegas_plot <-
  omegas_df |>
    mutate(Indicator = fct_relevel(Indicator,
                           "Extremes selection",
                           "Midpoint selection",
                           "Open question nonresponse",
                           "Closed question nonresponse",
                           "Weak nondiff (often)",
                           "Strong nondiff (sometimes)",
                           "Strong nondiff (often)",
                           "Break-off",
                           "Speeding"),
           LC   = fct_relevel(LC,   "Softsat", "Indifferent", "Extremist"),
           Mode = fct_relevel(Mode, "Paper", "Web")) |>
  group_by(Wave, Mode, Indicator) |>
  ggplot(aes(x = Indicator, y = ProbOmega, fill = Indicator, color = Indicator)) +
  geom_bar(stat = "identity", position = position_dodge2()) +

  geom_text(
    data = n_labels,
    aes(x = -Inf, y = Inf, label = label),
    inherit.aes = FALSE,
    hjust = -0.1, vjust = 1.5,
    size = 2.5,          # considerably smaller than 4.5
    color = "black"
  ) +

  facet_grid(rows = vars(Wave), cols = vars(LC, Mode), drop = TRUE) +

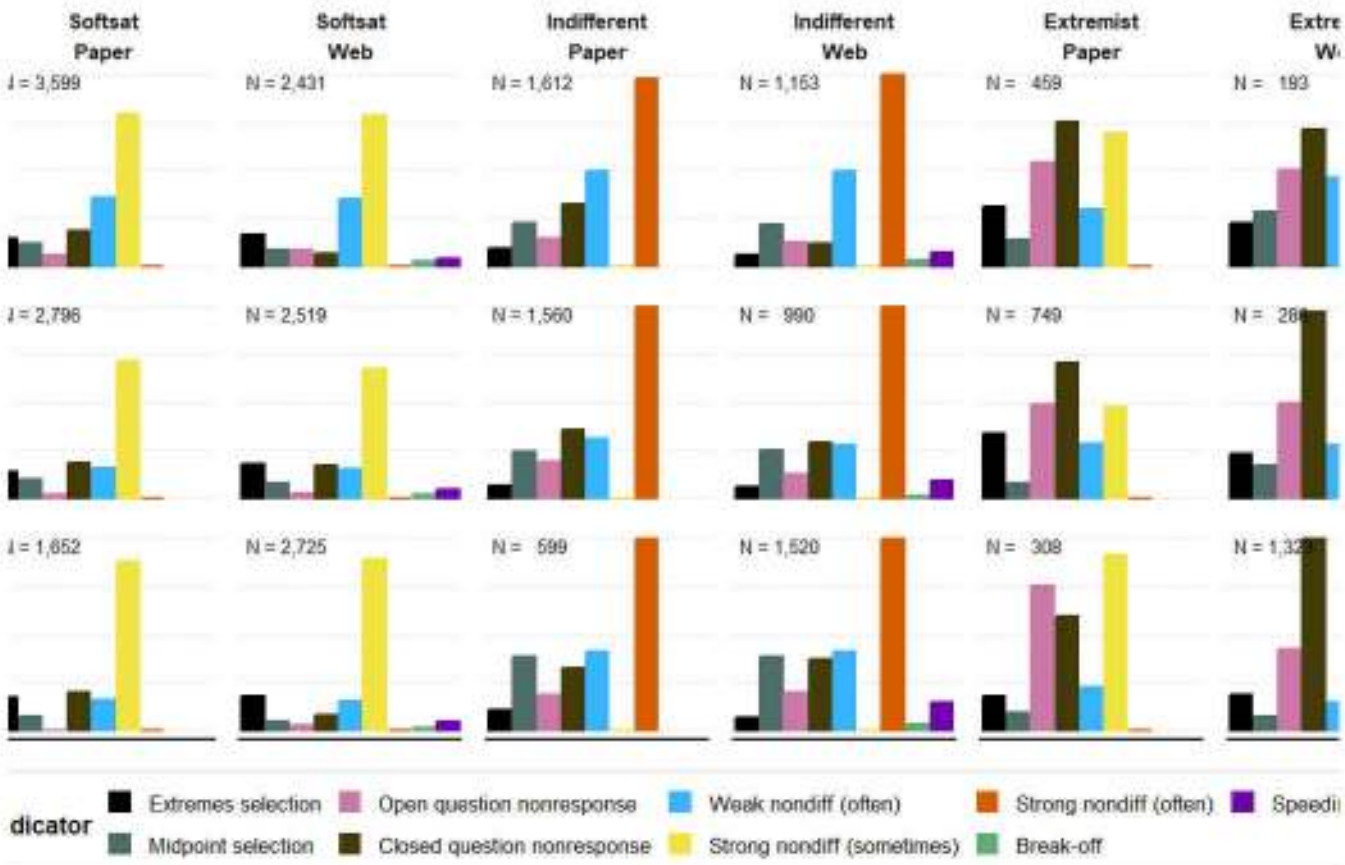
  force_panelsizes(
    rows = unit(rep(28, 3), "mm"),   # 3 Waves à 28 mm Höhe
    cols = unit(rep(30, 6), "mm")    # 6 LC/Mode-Kombis à 30 mm Breite
  ) +
```

```
  scale_fill_manual(values  = cbPalette) +
  scale_color_manual(values = cbPalette) +

  labs(y = expression(paste("Estimated Endorsement Probability (", hat(omega)[~italic("m|k")],
   ↪  ")"))) +

  papaja::theme_apa() +
  guides(
    fill  = guide_legend(
      nrow      = 2,
      keywidth  = unit(0.35, "cm"),
      keyheight = unit(0.35, "cm")
    ),
    color = "none"      # merge fill & color -> only one legend
  ) +
  theme(
    axis.text.x      = element_blank(),
    axis.title.x     = element_blank(),
    axis.ticks.x     = element_blank(),
    axis.text.y      = element_text(size = 8, color = "black"),
    axis.title.y     = element_text(family = "sans", size = 9, color = "black"),
    strip.text.x       = element_text(size = 8, face = "bold", margin = margin(t = 2, b = 2,
     ↪  unit = "pt")),
    strip.text.y     = element_text(size = 8, face = "bold", angle = -90),
    panel.spacing    = unit(0.3, "cm"),                  # enger als 0.8
    panel.grid.major.y = element_line(color = "grey", linewidth = 0.4, linetype = 3),
    legend.position      = "bottom",
    #legend.justification = "left",
    legend.location      = "plot",
    legend.title         = element_text(size = 9, face = "bold"),
    legend.text          = element_text(size = 8),       # runter von 12
    legend.box.background = element_rect(linetype = "solid", color = "lightgrey"),
    legend.box.margin    = margin(1, 1, 1, 1),
    legend.margin        = margin(2, 2, 2, 2, unit = "mm"),
    plot.margin          = margin(2, 2, 2, 2, unit = "mm")
  )

omegas_plot
```




```
# ggsave(file   = "Plots/omegas.png",
#        plot   = omegas_plot,
#        width  = 220,
#        height = 128,
#        units  = "mm",
#        dpi    = 300)
```

5.1 Omegas SensAn

```
omegas_df_sensan <- read_excel(paste0(userpath,
  "[redacted]/Sample_Summaries.xlsx"),
    sheet = "Omegas_SA") |>
  select(Wave:ProbOmega) |>
  mutate(
    LC = case_when(
      LC %in% "Complex" ~ "Extremist*",
      TRUE ~ LC
    )
  )


head(omegas_df_sensan,9)
```

```
# A tibble: 9 x 5
  Wave   Mode  LC         Indicator                ProbOmega
```

```
  <chr>  <chr> <chr>      <chr>                          <dbl>
1 Wave 1 Web   Extremist* Extremes selection             0.361
2 Wave 1 Web   Extremist* Midpoint selection             0.083
3 Wave 1 Web   Extremist* Open question nonresponse      0.057
4 Wave 1 Web   Extremist* Closed question nonresponse    0.158
5 Wave 1 Web   Extremist* Weak nondiff (often)           0.099
6 Wave 1 Web   Extremist* Strong nondiff (sometimes)     0
7 Wave 1 Web   Extremist* Strong nondiff (often)         0.159
8 Wave 1 Web   Extremist* Break-off                      0.576
9 Wave 1 Web   Extremist* Speeding                       0.122
```

```r
n_labels_sensan <-
  w1_sensan |>
  count(LC, Mode) |>
  filter(!is.na(LC), !is.na(Mode)) |>
  mutate(
    LC = case_when(
      LC %in% "Complex" ~ "Extremist*",
      TRUE ~ LC
    )
  ) |>
  mutate(
    label = paste0("N = ", format(n, big.mark = ",", scientific = FALSE)),
    LC   = fct_relevel(LC,   "Softsat", "Indifferent", "Extremist", "Extremist*"),
    Mode = fct_relevel(Mode, "Paper", "Web")
  )


cbPalette <- c("#000000", "#4F6E6A", "#CC79A7","#433C0A", "#39B4FF",
 ↪  "#F0E442","#D55E00","#61AD7B", "#7003A9" )

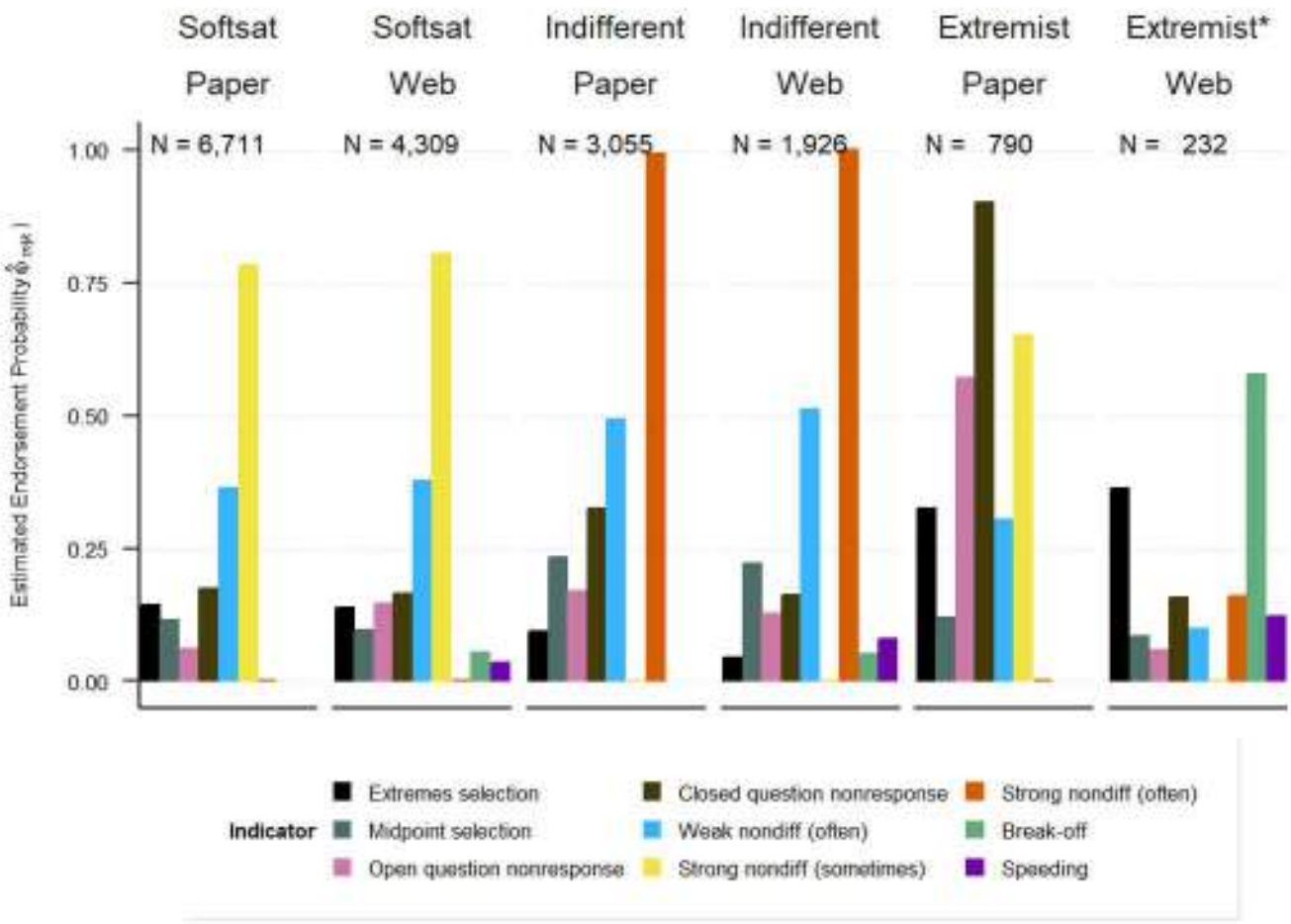
omegas_plot_sensan <-
  omegas_df_sensan |>
    mutate(Indicator = fct_relevel(Indicator,
                           "Extremes selection",
                           "Midpoint selection",
                           "Open question nonresponse",
                           "Closed question nonresponse",
                           "Weak nondiff (often)",
                           "Strong nondiff (sometimes)",
                           "Strong nondiff (often)",
                           "Break-off",
                           "Speeding"),
           LC = fct_relevel(LC,
                            "Softsat",
                            "Indifferent",
                            "Extremist",
                            "Extremist*"),
           Mode = fct_relevel(Mode,
                              "Paper",
                              "Web")) |>
  group_by(Mode, Indicator) |>
  ggplot(aes(x = Indicator, y = ProbOmega, fill = Indicator, color = Indicator)) +
  geom_bar(stat = "identity",
           position=position_dodge2()) +
```

```
  geom_text(
    data = n_labels_sensan,
    aes(x = -Inf, y = Inf, label = label),
    inherit.aes = FALSE,
    hjust = -0.1,
    vjust = 1.5,
    size = 3.5,
    color = "black"
  ) +

  facet_grid(#rows = vars(Wave),
             cols = vars(LC, Mode), drop = TRUE)+ #+ coord_flip()
  scale_fill_manual(values=cbPalette) +
  labs (x = "",
        y = expression(paste("Estimated Endorsement Probability (", hat(omega)[~italic("m|k")],
          ↪  ")"))) +


  scale_color_manual(values=cbPalette) +
  papaja::theme_apa() +
    # guides(color = guide_legend(override.aes = list(size = 0.2))) +
  theme(legend.position = "bottom",
       # axis.text.x = element_text(angle = 30, family = "arial", hjust = 1, size = 11, color =
         ↪  "black"),
       axis.text.x = element_blank(),
        axis.title.x = element_blank(),
       axis.ticks.x = element_blank(),
       axis.text.y = element_text(size = 8, color = "black"),
        axis.title.y = element_text(family = "sans", size = 8, color = "black"),
         legend.title = element_text(size = 8, face = "bold"),
              legend.text  = element_text(size = 8),
        strip.text = element_text(size = 10),
        panel.spacing = unit(0.2, "cm", data = NULL),
        panel.grid.major.y = element_line(color = "grey",
                                          linewidth = 0.4,
                                          linetype = 3),
       legend.box.background = element_rect(linetype = "solid", color = "lightgrey")) +
  guides(
  fill  = guide_legend(
    title         = "Indicator",
    nrow          = 3,                          # 2 rows for the 9 items
    keywidth      = unit(0.3, "cm"),            # Kästchen schmaler
    keyheight     = unit(0.3, "cm"),            # Kästchen flacher
    override.aes  = list(size = 0.2)
  ),
  color = "none"                                # doppelte Legende (fill+color) unterdrücken
)

omegas_plot_sensan
```




```
# ggsave(file = "Plots/omegas_sensan.png", omegas_plot_sensan,
#        width = 165.1, height = 100, units = "mm", dpi = 300)
#
    # width  = 297,
    #   height = 0.5*220,
    #   units  = "mm",
    #   dpi    = 300)
```

6 Barplot Global

```
p_load("readxl")

global_df <- read_excel(paste0(userpath,
  "[redacted]/Sample_Summaries.xlsx"),
    sheet = "GlobalProp") |>
 select(Model:ProbMostLikely) |>
 mutate(Model = str_replace(Model, "_", "\n")) |>
 mutate(Model = fct_relevel(Model,
                            "Wave 1 Paper\n (N = 5,670)",
                            "Wave 1 Web\n (N = 3,777)",
                            "Wave 2 Paper\n (N = 5,105)",
                            "Wave 2 Web\n (N = 3,795)",
                            "Wave 3 Paper\n (N = 2,559)",
                            "Wave 3 Web\n (N = 5,568)"))
```

```
head(global_df,9)
```

```
# A tibble: 9 x 4
  Model                       Pattern     PropPostProb ProbMostLikely
  <fct>                       <chr>              <dbl>          <dbl>
1 "Wave 1 Web\n (N = 3,777)" Softsat           0.613          0.644
2 "Wave 1 Web\n (N = 3,777)" Extremist         0.0817         0.0511
3 "Wave 1 Web\n (N = 3,777)" Indifferent       0.305          0.305
4 "Wave 2 Web\n (N = 3,795)" Softsat           0.623          0.664
5 "Wave 2 Web\n (N = 3,795)" Extremist         0.116          0.0754
6 "Wave 2 Web\n (N = 3,795)" Indifferent       0.261          0.261
7 "Wave 3 Web\n (N = 5,568)" Softsat           0.531          0.489
8 "Wave 3 Web\n (N = 5,568)" Extremist         0.196          0.238
9 "Wave 3 Web\n (N = 5,568)" Indifferent       0.273          0.273
```

```
global_df_sensan <- read_excel(paste0(userpath,
  "[redacted]/Sample_Summaries.xlsx"),
     sheet = "GlobalProp_SA") |>
  select(Model:ProbMostLikely) |>
  slice(1:6) |>
  mutate(Model = str_replace(Model, "_", "\n")) |>
  mutate(Model = fct_relevel(Model,
                             "Paper\n (N = 10,556)",
                             "Web\n (N = 6,467)"),
         Pattern = case_when(Pattern %in% c("Complex", "Extremist") -"Extremist*",
                             TRUE -Pattern))


head(global_df_sensan,9)
```

```
# A tibble: 6 x 4
  Model                   Pattern     PropPostProb ProbMostLikely
  <fct>                   <chr>              <dbl>          <dbl>
1 "Web\n (N = 6,467)"     Softsat           0.648          0.666
2 "Web\n (N = 6,467)"     Extremist*        0.0617         0.0359
3 "Web\n (N = 6,467)"     Indifferent       0.29           0.298
4 "Paper\n (N = 10,556)" Softsat           0.590          0.636
5 "Paper\n (N = 10,556)" Extremist*        0.118          0.0748
6 "Paper\n (N = 10,556)" Indifferent       0.292          0.289
```

```
# plain labels first (no plotmath expressions)
global_df_long <- global_df |>
  pivot_longer(
    cols = c(PropPostProb, ProbMostLikely),
    names_to  = "ProbType",
    values_to = "Value"
  ) |>
  mutate(ProbType = factor(ProbType,
                           levels = c("PropPostProb", "ProbMostLikely")))
                           # no labels are applied in this step

# Labeller-Funktion separat definieren
prob_labeller <- as_labeller(
  c(PropPostProb   = "A: pi_hat",
    ProbMostLikely = "B: macP_c"),
```

```
  default = label_parsed   # <-- parsed-Modus
)

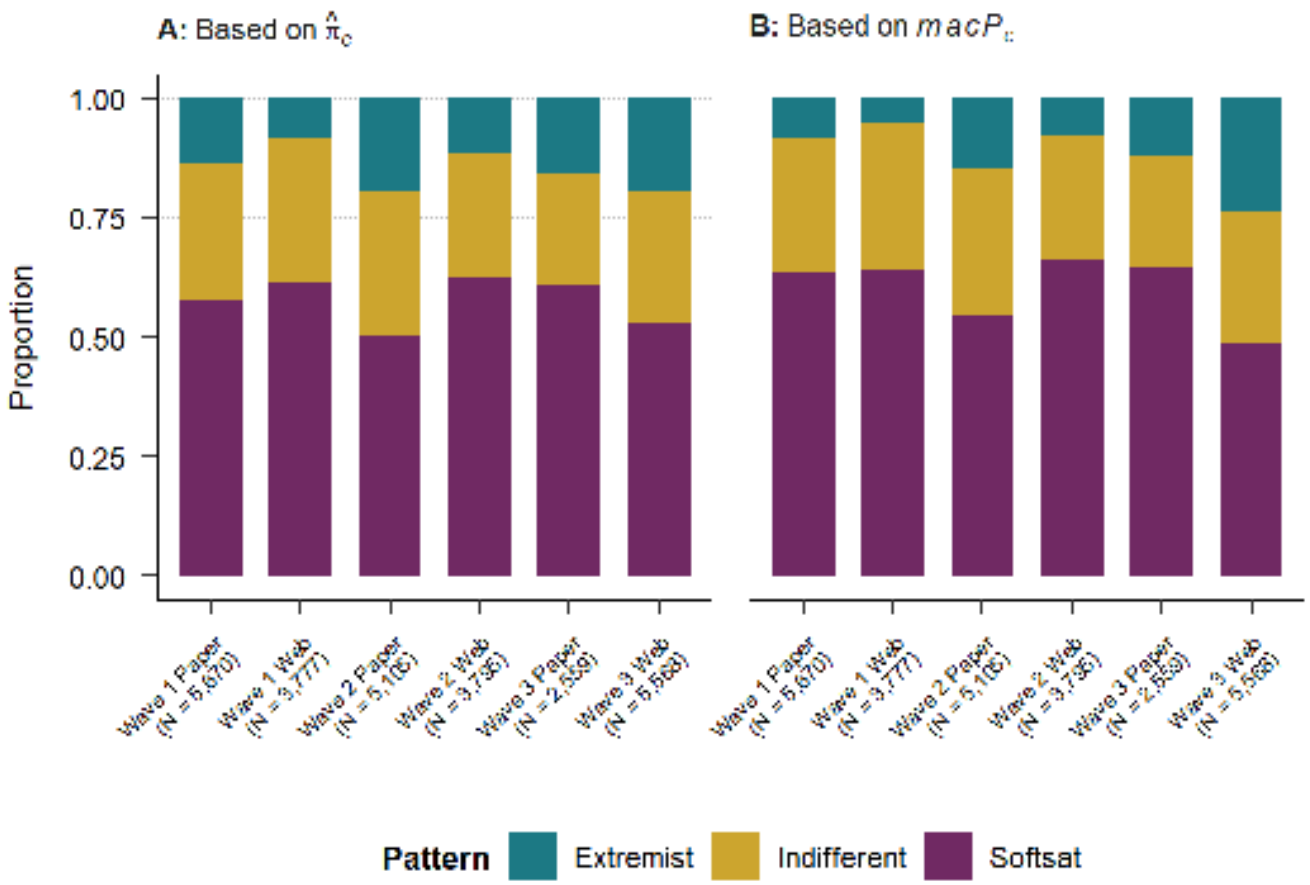
# for parsed labels, plotmath syntax is passed as a character string:
prob_labeller <- as_labeller(
  c(PropPostProb   = "bold('A:')~'Based on'~hat(pi)[italic('c')]",
    ProbMostLikely = "bold('B:')~'Based on'~italic(macP)[italic('c')]"),
  default = label_parsed
)

global_grid <- global_df_long |>
  ggplot(aes(x = Model, y = Value, fill = Pattern)) +
  geom_bar(stat = "identity",
           position = position_stack(),
           width = 0.7) +
  scale_fill_manual(
    values = c("#1C7984", "#CCA52E", "#702963"),
    labels = c("Extremist", "Indifferent", "Softsat")
  ) +

facet_grid(
  . ~ ProbType,
  scales  = "fixed",
  labeller = prob_labeller
) +

  labs(y = "Proportion") +
  papaja::theme_apa() +
  theme(
    axis.text.x  = element_text(angle = 45, hjust = 1, vjust = 1,
                                family = "sans", size = 8, color = "black"),
    axis.text.y  = element_text(color = "black"),
    axis.title.x = element_blank(),
    axis.title.y = element_text(color = "black", size = 12),
    panel.grid.major.y = element_line(color = "grey", linewidth = 0.4, linetype = 3),
    strip.text   = element_text(size = 9, hjust = 0),  # Panel-Labels "A" / "B"
    legend.position = "bottom",
    legend.title = element_text(size = 12, face = "bold" ),
    legend.text  = element_text(size = 11)
  )

global_grid
```




```
#ggsave(file = "Plots/global_grid.png", global_grid,
#       width = 165.1, height = 110, units = "mm", dpi = 300)
```

7 Barplot Sensan

```
# plain labels first (no plotmath expressions)
global_df_sensan_long <- global_df_sensan |>
  pivot_longer(
    cols = c(PropPostProb, ProbMostLikely),
    names_to  = "ProbType",
    values_to = "Value"
  ) |>
  mutate(ProbType = factor(ProbType,
                           levels = c("PropPostProb", "ProbMostLikely")))
                           # no labels are applied in this step

# Labeller-Funktion separat definieren
prob_labeller <- as_labeller(
  c(PropPostProb   = "A: pi_hat",
    ProbMostLikely = "B: macP_c"),
  default = label_parsed   # <-- parsed-Modus
)

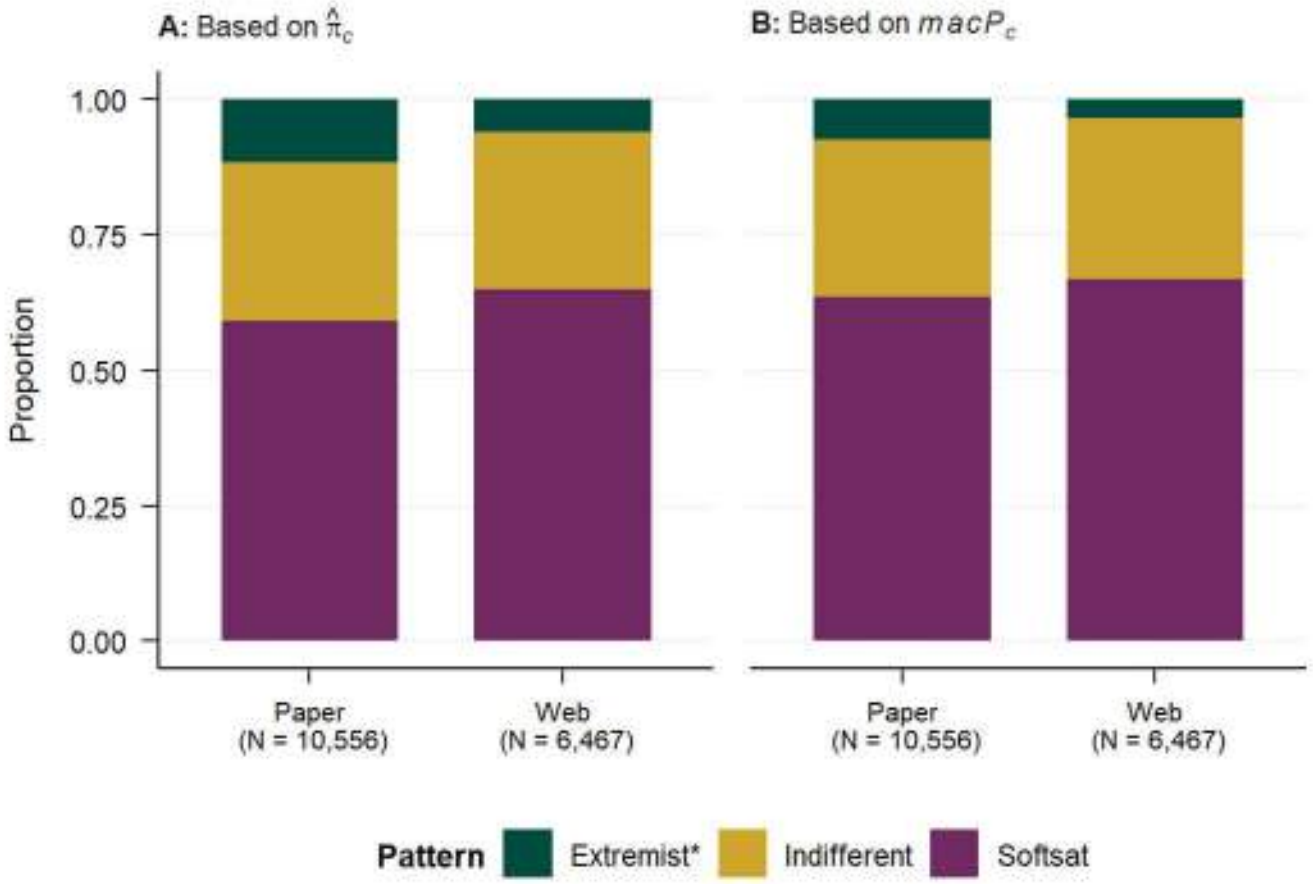
# for parsed labels, plotmath syntax is passed as a character string:
```

```r
prob_labeller <- as_labeller(
  c(PropPostProb   = "bold('A:')~'Based on'~hat(pi)[italic('c')]",
    ProbMostLikely = "bold('B:')~'Based on'~italic(macP)[italic('c')]"),
  default = label_parsed
)

global_grid_sensan <-
  global_df_sensan_long |>
  ggplot(aes(x = Model, y = Value, fill = Pattern)) +
  geom_bar(stat = "identity",
           position = position_stack(),
           width = 0.7) +
  scale_fill_manual(
    values = c("#004D40", "#CCA52E", "#702963"),
    labels = c("Extremist*", "Indifferent", "Softsat")
  ) +

facet_grid(
  . ~ ProbType,
  scales  = "fixed",
  labeller = prob_labeller
) +
  labs(y = "Proportion") +
  papaja::theme_apa() +
  theme(
    axis.text.x  = element_text(
                                family = "sans", size = 10, color = "black"),
    axis.text.y  = element_text(color = "black"),
    axis.title.x = element_blank(),
    axis.title.y = element_text(color = "black", size = 12),
    panel.grid.major.y = element_line(color = "grey", linewidth = 0.4, linetype = 3),
    strip.text   = element_text(size = 9, hjust = 0),  # Panel-Labels "A" / "B"
    legend.position = "bottom",
    legend.title = element_text(size = 12, face = "bold"),
    legend.text  = element_text(size = 11)
  )

global_grid_sensan
```




```
#ggsave(file = "Plots/global_grid_sensan.png", global_grid_sensan,
#       width = 140.1, height = 110, units = "mm", dpi = 300)
```

8 Multinoms SocDem

8.1 Sample

```
p_load("readxl")

mn_df <-
  read_excel(paste0(userpath,
    "[redacted]/Sample_Summaries.xlsx"),
     sheet = "MultinomSocDem") |>
  select(Model:UUOR) |>
  mutate(Model = str_replace(Model, "_", "\n")) |>
     #    Variable = str_replace(Variable, "_", "\n")) |>
    mutate(Model = fct_relevel(Model,
                               "Wave 1 Paper\n (N = 5,670)",
                               "Wave 1 Web\n (N = 3,777)",
                               "Wave 2 Paper\n (N = 5,105)",
                               "Wave 2 Web\n (N = 3,795)",
                               "Wave 3 Paper\n (N = 2,559)",
                               "Wave 3 Web\n (N = 5,568)")) |>

  mutate(
```

```
    # clean NAs and infinite values
    across(c(`Est (log-Odds)`, LLLogit, UUlogit),
           ~ if_else(is.infinite(.x), NA_real_, .x)),

    # significance is evaluated on the logit scale
    significant = case_when(
      LLLogit > 0 & UUlogit > 0 ~ "positive",
      LLLogit < 0 & UUlogit < 0 ~ "negative",
      TRUE                      ~ "insignificant"
    ),

    # flags marking estimates outside the plotting range
    clipped_high = is.na(`Est (log-Odds)`) | UUlogit > 2 | `Est (log-Odds)` > 2,
    clipped_low  = is.na(`Est (log-Odds)`) | LLLogit < -2 | `Est (log-Odds)` < -2,
    clipped      = clipped_high | clipped_low,

    # plotted values (clipped on both sides)
    Est_plot = pmin(pmax(`Est (log-Odds)`, -2), 2),
    LL_plot  = pmin(pmax(LLLogit, -2), 2),
    UU_plot  = pmin(pmax(UUlogit, -2), 2),

    # arrow start points (minimum shaft length of 0.3 units)
    arrow_start_high = pmin(LL_plot,  1.7),
    arrow_start_low  = pmax(UU_plot, -1.7)
  )


head(mn_df,9)
```

```
# A tibble: 9 x 21
  Model               Variable Comparison `Est (log-Odds)`  S.E. `Est./S.E. (z)`
  <fct>               <chr>    <chr>                 <dbl> <dbl>           <dbl>
1 "Wave 1 Web\n (N =~ Househo~ Extremist~            0.67  0.359           1.87
2 "Wave 1 Web\n (N =~ Age: Yo~ Extremist~           -0.506 0.37           -1.37
3 "Wave 1 Web\n (N =~ Age: Ol~ Extremist~            0.405 0.706           0.574
4 "Wave 1 Web\n (N =~ Gender:~ Extremist~            0.497 0.306           1.62
5 "Wave 1 Web\n (N =~ Migrati~ Extremist~            0.178 0.36            0.494
6 "Wave 1 Web\n (N =~ Educati~ Extremist~            0.259 0.301           0.861
7 "Wave 1 Web\n (N =~ Educati~ Extremist~            0.115 1.47            0.078
8 "Wave 1 Web\n (N =~ Employm~ Extremist~           -0.088 0.404          -0.219
9 "Wave 1 Web\n (N =~ Employm~ Extremist~            0.194 0.39            0.497
# i 15 more variables: `Two-Tailed p-value` <dbl>, OR <dbl>, LLLogit <dbl>,
#   LLOR <dbl>, UUlogit <dbl>, UUOR <dbl>, significant <chr>,
#   clipped_high <lgl>, clipped_low <lgl>, clipped <lgl>, Est_plot <dbl>,
#   LL_plot <dbl>, UU_plot <dbl>, arrow_start_high <dbl>, arrow_start_low <dbl>
```

```
posn_d2 <- position_dodge(1)


#p_load(ggh4x)

multinom <-
  mn_df |>
  ggplot(aes(x     = Est_plot,
             y     = as.factor(Variable),
             group = Model,
```

```
            color = significant)) +

  geom_point(data = ~filter(., !clipped),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_errorbarh(data = ~filter(., !clipped),
                 aes(xmin = LL_plot, xmax = UU_plot),
                 position = posn_d2) +
  geom_point(data = ~filter(., clipped_high & !clipped_low),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_segment(data = ~filter(., clipped_high & !clipped_low),
               aes(x = arrow_start_high, xend = 2,
                   y = as.factor(Variable), yend = as.factor(Variable)),
               arrow     = arrow(length = unit(0.2, "cm"), type = "closed"),
               linewidth = 0.6,
               position  = posn_d2,
               show.legend = FALSE) +
  geom_point(data = ~filter(., clipped_low & !clipped_high),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_segment(data = ~filter(., clipped_low & !clipped_high),
               aes(x = arrow_start_low, xend = -2,
                   y = as.factor(Variable), yend = as.factor(Variable)),
               arrow     = arrow(length = unit(0.2, "cm"), type = "closed"),
               linewidth = 0.6,
               position  = posn_d2,
               show.legend = FALSE) +

  geom_vline(xintercept = 0, color = "grey", linetype = "dotted") +

  scale_color_manual(
    values = c("negative"      = "#1A85FF",
               "positive"      = "#D41159",
               "insignificant" = "#4D4E4E"),
    labels = c("negative"      = "Negative association (CI < 0)",
               "positive"      = "Positive association (CI > 0)",
               "insignificant" = "Non-significant (CI includes 0)"),
    name  = NULL,
    guide = guide_legend(override.aes = list(linewidth = 1, shape = 19))
  ) +

  coord_cartesian(xlim = c(-2, 2), clip = "off") +
  scale_x_continuous(breaks = seq(-2, 2, by = 1)) +

  labs(x = "Log-Odds with 95%-CI", y = NULL) +

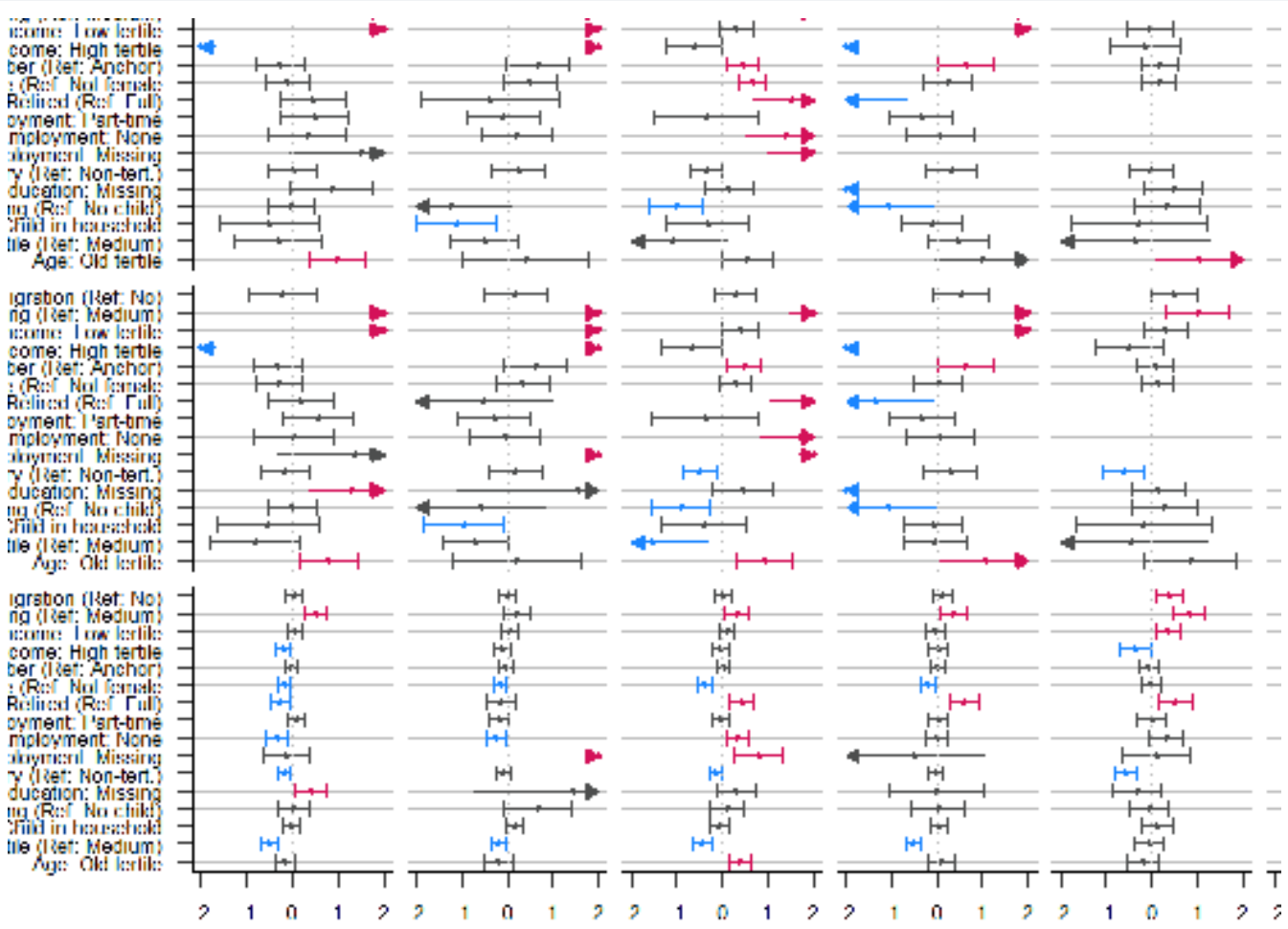
  facet_grid(rows = vars(Comparison),
             cols = vars(Model),
             drop = TRUE) +

  force_panelsizes(
    rows = unit(rep(40, 3), "mm"),   # jedes der 3 Row-Panels: 40 mm hoch
```

```
    cols = unit(rep(28, 6), "mm")    # jedes der 6 Col-Panels: 28 mm breit
  ) +

  papaja::theme_apa() +
  theme(
    axis.title.x    = element_text(size = 8, margin = margin(t = 5)),
    axis.text.x     = element_text(size = 8, color = "black"),
    axis.text.y     = element_text(size = 8, color = "black"),
    strip.text.x    = element_text(size = 9, margin = margin(b = 3, t = 3)),
    strip.text.y    = element_text(size = 9, angle = -90, margin = margin(l = 2, r = 2)),
    axis.title.y    = element_blank(),
    panel.grid.major.y   = element_line(color = "grey", linewidth = 0.1),
    legend.position      = "bottom",
    legend.margin        = margin(t = 8),
    legend.box.spacing   = unit(0, "pt"),
    legend.text          = element_text(size = 8),
    panel.spacing        = unit(0.2, "cm"),
    panel.background     = element_rect(fill = "white"),
    strip.background     = element_rect(fill = "grey92", color = NA),
    plot.margin = margin(t = 0, r = 0, b = 0, l = 0, unit = "mm")
  )

multinom
```




```
# ggsave(file   = "Plots/multinom.png",
```

```
#       plot   = multinom,
#       width  = 240,
#       height = 155,
#       units  = "mm",
#       dpi    = 300,
#       bg = "transparent")
```

8.2 SensAn

```
p_load("readxl")

mn_df_sensan <-
  read_excel(paste0(userpath,
    "[redacted]/Sample_Summaries.xlsx"),
     sheet = "MultinomSocDem_SA") |>
  select(Model:UUOR) |>
  mutate(Model = str_replace(Model, "_", "\n")) |>
     #    Variable = str_replace(Variable, "_", "\n")) |>
    mutate(Model = fct_relevel(Model,
                               "Paper\n (N = 10,556)",
                               "Web\n (N = 6,467)")) |>

  mutate(
    # clean NAs and infinite values
    across(c(`Est (log-Odds)`, LLLogit, UUlogit),
           ~ if_else(is.infinite(.x), NA_real_, .x)),

    # significance is evaluated on the logit scale
    significant = case_when(
      LLLogit > 0 & UUlogit > 0 ~ "positive",
      LLLogit < 0 & UUlogit < 0 ~ "negative",
      TRUE                      ~ "insignificant"
    ),

    # flags marking estimates outside the plotting range
    clipped_high = is.na(`Est (log-Odds)`) | UUlogit > 2 | `Est (log-Odds)` > 2,
    clipped_low  = is.na(`Est (log-Odds)`) | LLLogit < -2 | `Est (log-Odds)` < -2,
    clipped      = clipped_high | clipped_low,

    # plotted values (clipped on both sides)
    Est_plot = pmin(pmax(`Est (log-Odds)`, -2), 2),
    LL_plot  = pmin(pmax(LLLogit, -2), 2),
    UU_plot  = pmin(pmax(UUlogit, -2), 2),

    # arrow start points (minimum shaft length of 0.3 units)
    arrow_start_high = pmin(LL_plot,  1.7),
    arrow_start_low  = pmax(UU_plot, -1.7)
  ) |>

  mutate(
  Comparison = case_when(
    Comparison %in% c("Extremist* vs. Indifferent") ~ "Extremist*\n vs. Indifferent",
    Comparison %in% c("Extremist* vs. Softsat")     ~ "Extremist*\n vs Softsat",
    Comparison %in% "Indifferent vs. Softsat"       ~ "Indifferent\n vs. Softsat"
  ),
```

```
  # set the category order explicitly:
  Comparison = fct_relevel(Comparison,
    "Extremist*\n vs. Indifferent",
    "Extremist*\n vs Softsat",
    "Indifferent\n vs. Softsat"
  )
)




multinom_sensan <-
  mn_df_sensan |>
  ggplot(aes(x     = Est_plot,
             y     = as.factor(Variable),
             group = Model,
             color = significant)) +

  geom_point(data = ~filter(., !clipped),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_errorbarh(data = ~filter(., !clipped),
                 aes(xmin = LL_plot, xmax = UU_plot),
                 position = posn_d2) +
  geom_point(data = ~filter(., clipped_high & !clipped_low),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_segment(data = ~filter(., clipped_high & !clipped_low),
               aes(x = arrow_start_high, xend = 2,
                   y = as.factor(Variable), yend = as.factor(Variable)),
               arrow     = arrow(length = unit(0.2, "cm"), type = "closed"),
               linewidth = 0.6,
               position  = posn_d2,
               show.legend = FALSE) +
  geom_point(data = ~filter(., clipped_low & !clipped_high),
             aes(x = Est_plot),
             size = 1,
             position = posn_d2) +
  geom_segment(data = ~filter(., clipped_low & !clipped_high),
               aes(x = arrow_start_low, xend = -2,
                   y = as.factor(Variable), yend = as.factor(Variable)),
               arrow     = arrow(length = unit(0.2, "cm"), type = "closed"),
               linewidth = 0.6,
               position  = posn_d2,
               show.legend = FALSE) +

  geom_vline(xintercept = 0, color = "grey", linetype = "dotted") +

  scale_color_manual(
    values = c("negative"      = "#1A85FF",
               "positive"      = "#D41159",
               "insignificant" = "#4D4E4E"),
    labels = c("negative"      = "Negative (CI < 0)",
               "positive"      = "Positive (CI > 0)",
```

```
              "insignificant" = "Non-significant (CI includes 0)"),
    name  = "Association",
    guide = guide_legend(override.aes = list(linewidth = 1, shape = 19),
                         title = "Association",
                         nrow = 1)
  ) +

  coord_cartesian(xlim = c(-2, 2), clip = "off") +
  scale_x_continuous(breaks = seq(-2, 2, by = 1)) +

  labs(x = "Log-Odds with 95%-CI", y = NULL) +

  facet_grid(rows = vars(Comparison),
             cols = vars(Model),
             drop = TRUE) +

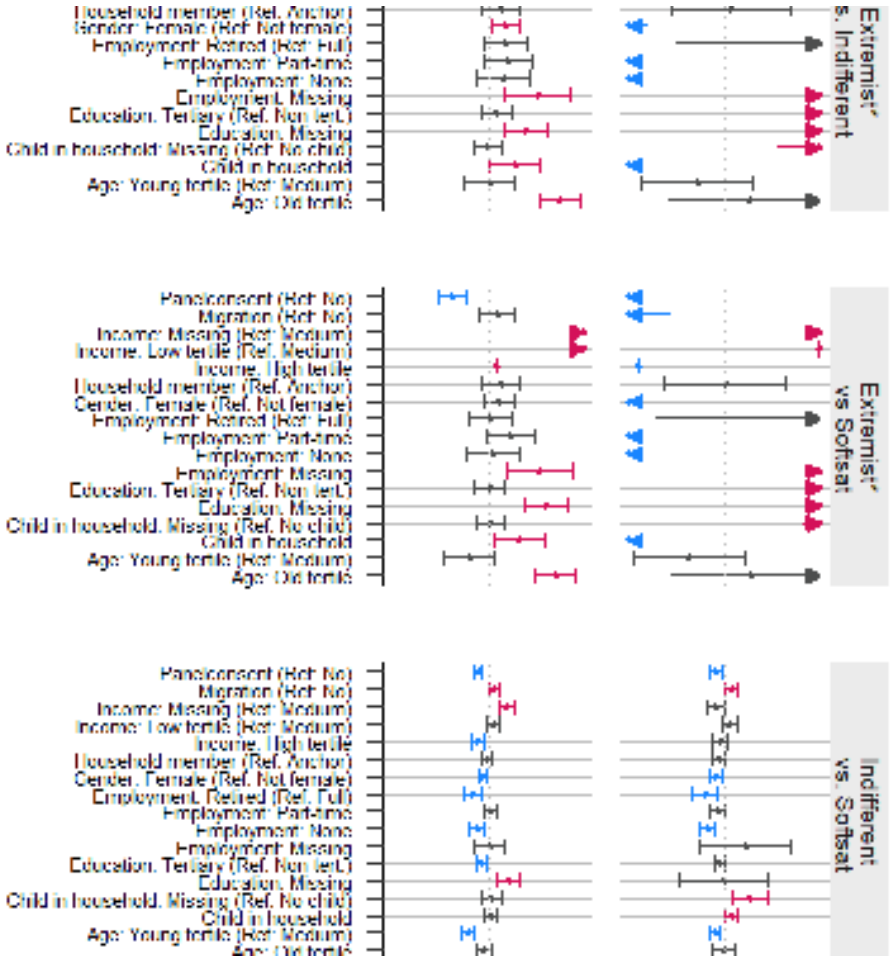
  force_panelsizes(
    rows = unit(rep(40, 3), "mm"),   # 3 Comparison-Rows à 40 mm
    cols = unit(rep(28, 3), "mm")    # ggf. Spaltenanzahl anpassen
  ) +

  papaja::theme_apa() +
  theme(
    axis.title.x    = element_text(size = 8, margin = margin(t = 5)),
    axis.text.x     = element_text(size = 8, color = "black"),
    axis.text.y     = element_text(size = 7, color = "black"),
    strip.text.x    = element_text(size = 9, margin = margin(b = 3, t = 3)),
    strip.text.y    = element_text(size = 9, angle = -90, margin = margin(l = 2, r = 2)),
    axis.title.y    = element_blank(),
    panel.grid.major.y   = element_line(color = "grey", linewidth = 0.1),

    legend.title = element_text(size = 8,face = "bold"),
    legend.position      = "bottom",
    legend.justification = "left",    # <-- linksbündig
    legend.location      = "plot",    # <-- relativ zum gesamten Plot, nicht nur Panel
    legend.box.margin = margin(l = 5, unit = "mm"),
    legend.margin        = margin(t = 2, l = 5, unit = "mm"),
    legend.box.spacing   = unit(0, "pt"),
    plot.margin = margin(t = 0, r = 0, b = 0, l = 0, unit = "mm"),

    legend.text          = element_text(size = 8),
    panel.spacing.x      = unit(0.7, "lines"),
    panel.spacing.y      = unit(2, "lines"),
    panel.background     = element_rect(fill = "white"),
    strip.background     = element_rect(fill = "grey92", color = NA)
  )

multinom_sensan
```




```
# ggsave(file  = "Plots/multinom_sensan.png",
#        plot  = multinom_sensan,
#        width  = 165,    # usable width of an A4 portrait page
#        height = 165,    # height limited to an A4 portrait page incl. margins
#        units  = "mm",
#        dpi    = 300)
```